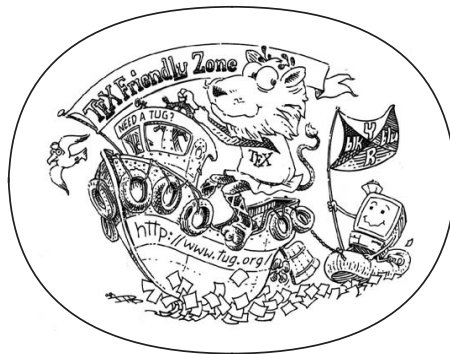


ANDRÉ MIEDE & IVO PLETIKOSIĆ
A CLASSIC THESIS STYLE

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ANDRÉ MIEDE & IVO PLETIKOSIĆ



An Homage to The Elements of Typographic Style

February 2025 – classicthesis v4.8

André Miede & Ivo Pletikosić: *A Classic Thesis Style*, An Homage to
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Ohana means family.
Family means nobody gets left behind, or forgotten.
— Lilo & Stitch

Dedicated to the loving memory of Rudolf Miede.
1939–2005

ABSTRACT

Short summary of the contents in English... a great guide by Kent Beck how to write good abstracts can be found here:

<https://plg.uwaterloo.ca/~migod/research/beck00PSLA.html>

ZUSAMMENFASSUNG

Kurze Zusammenfassung des Inhaltes in deutscher Sprache...

PUBLICATIONS

This might come in handy for PhD theses: some ideas and figures have appeared previously in the following publications:

This is just an example.

- [1] Tobias Isenberg, André Miede, and Sheelagh Carpendale. “A Buffer Framework for Supporting Responsive Interaction in Information Visualization Interfaces.” In: *Proceedings of the Fourth International Conference on Creating, Connecting, and Collaborating through Computing (C⁵ 2006)*. IEEE, 2006, pp. 262–269. ISBN: 978-0-7695-2563-1.
- [2] Ulrich Lampe, Markus Kieselmann, André Miede, Sebastian Zöller, and Ralf Steinmetz. “A Tale of Millis and Nanos: On the Accuracy of Time Measurements in Virtual Machines.” In: *Proceedings of the Second European Conference on Service-Oriented and Cloud Computing (ESOCC 2013)*. Springer, 2013, pp. 172–179. ISBN: 978-3-642-40650-8.
- [3] Ulrich Lampe, Qiong Wu, Ronny Hans, André Miede, and Ralf Steinmetz. “To Frag Or To Be Fraggd – An Empirical Assessment of Latency in Cloud Gaming.” In: *Proceedings of the Third International Conference on Cloud Computing and Services Science (CLOSER 2013)*. 2013, pp. 5–12. ISBN: 978-898-8565-52-5.
- [4] André Miede. “Theses and other Beautiful Documents with classicthesis.” In: *TUGboat – The Communications of the T_EX Users Group* 31.1 (2010), pp. 18–20. ISSN: 0896-3207.
- [5] André Miede, Gökhan Şimşek, Stefan Schulte, Daniel F. Abawi, Julian Eckert, and Ralf Steinmetz. “Revealing Business Relationships – Eavesdropping Cross-organizational Collaboration in the Internet of Services.” In: *Proceedings of the Tenth International Conference Wirtschaftsinformatik (WI 2011)*. Vol. 2. 2011, pp. 1083–1092. ISBN: 978-1-4467-9236-0.
- [6] Hsin-Yi Tsai, Melanie Siebenhaar, André Miede, Yu-Lun Huang, and Ralf Steinmetz. “Threat as a Service? Virtualization’s Impact on Cloud Security.” In: *IEEE IT Professional* 14.1 (2012), pp. 32–37. ISSN: 1520-9202.

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*We have seen that computer programming is an art,
because it applies accumulated knowledge to the world,
because it requires skill and ingenuity, and especially
because it produces objects of beauty.*

— Donald E. Knuth [61]

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Put your acknowledgments here.

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ACRONYMS

Part I

SOME KIND OF MANUAL

EFFICIENT LANGUAGE MODELS

1.1 MODELING THE LANGUAGE

1.1.1 *A dialog between Shannon and Chomsky*

It is curious that we still don't know where the expression *language model* come from and when it establishes. Not the notion of what is a *language model* but the expression itself. Because for the notion, we know that the idea of language model is quite old and one of the earliest formulations closely related to what we now call language modeling can be found in Shannon's work on information theory Shannon [94]. Shannon starts from a simple observation: sequences of linguistic symbols are not random, but exhibit regularities characteristic of the language.

"[...] the messages to be transmitted consist of sequences of letters. These sequences, however, are not completely random. In general, they form sentences and have the statistical structure of, say, English."

— Claude E. Shannon, *A Mathematical Theory of Communication*
(1948)

If English sentences have a statistical structure, then a statistical model that captures this structure should be able to generate sequences that look like English. But what is this statistical model? How can we estimate it? Later on, again Shannon gives a more operational view:

"We can think of a discrete source as generating the message, symbol by symbol. It will choose successive symbols according to certain probabilities depending, in general, on preceding choices [...]."

— Claude E. Shannon, *A Mathematical Theory of Communication*
(1948)

Shannon describes text generation as a sequential probabilistic process: the next symbol is sampled according to a probability distribution that depends on the preceding context. He then considers increasingly higher-order approximations, where the next symbol depends on a progressively longer context. These probabilities can be estimated from text by simply estimating how frequently a symbol occurs after a given context. This already contains the basic ingredients of what we would now recognize as an autoregressive language model: modeling the probability of the next symbol conditioned on its preceding context.

We will discuss this more formally. Still, while Shannon describes what conceptually is the language model as we perceive it today, he doesn't use the expression *language model*. Calling these systems as such could suggest that they are a linguistic model of a language. But does a better statistical model of English necessarily mean a better model of its grammar? While this idea is natural from a corpus-based view of language, it is strongly rejected by Chomsky. In *Three Models for the Description of Language*, he states Chomsky [22, p. 113]:

"We find that no finite-state Markov process that produces symbols with transition from state to state can serve as an English grammar. [...] such processes that produce n-order statistical approximations to English do not come closer, with increasing n, to matching the output of an English grammar."

— Noam Chomsky, *Three Models for the Description of Language*
(1956)

There are two main claims made by Chomsky in this paper. The first claim is structural and concerns the **bounded memory** of finite-state models. For finite-order Markov language models, this limitation manifests as a fixed context length: only a bounded portion of the preceding sequence can determine the next symbol. English, however, permits recursive constructions whose dependencies can require information from an arbitrarily large preceding structure. Modern language models have greatly increased the amount of preceding context they can exploit, making them much better at modeling long-range dependencies. Yet practical models still operate with bounded context and memory, so the distinction highlighted by Chomsky between unbounded linguistic structure and bounded computational memory remains relevant. Second, even if we consider a statistical model with unbounded memory, Chomsky's criticism goes beyond model capacity. His broader claim is that **language innate**, biologically programmed in the human brain Chomsky [23] and cannot be learned simply by accumulating statistics over observed linguistic data. The main argument is the *poverty of stimulus*: a child learns its native language rapidly, despite being exposed to limited, noisy, and incomplete examples. From a Chomskyan perspective, even highly fluent language models cannot be said, on behavioral evidence alone, to have acquired the human language faculty. Their success may demonstrate powerful statistical learning from linguistic experience, but Chomsky's theory holds that human linguistic competence depends on internal constraints that cannot be inferred from experience alone.

So it is difficult to pinpoint when the expression *language modeling* became established, and what exactly was originally meant by it. From a Chomskian perspective, a model of a language should explain the structure of that language; Chomsky explicitly rejected the idea that a statistical model fitted to linguistic data could constitute such a

model. Shannon, meanwhile, developed statistical models of English but, importantly, did not call them *language models*. The story takes a different turn in the early 1970s. In 1974, a team at IBM working on speech recognition described a statistical model closely related to Shannon’s formulation and explicitly called it a *language model*. In their probabilistic decomposition of speech recognition, the language model is simply the component that assigns probabilities to possible word sequences—that is, a model of which utterances are more or less likely in a given language. As they put it, “it is the function of the language model (LM) to provide us with $\Pr\{W \mid L\}$,” where W denotes a word sequence and L the language. This work is therefore probably one of the earliest uses of the expression *language model* in essentially its modern statistical sense: a probability distribution over linguistic sequences estimated from language data.

Indeed, this is not a model of a language in the Chomskyan sense, since language is assumed to rely on innate structure and therefore cannot be modeled from experience alone. It is not straightforward, however, to translate this notion of innateness into the language of machine learning. Does Chomsky’s argument imply that general-purpose learning architectures are inherently incapable of acquiring language from data alone? Or can we instead design computational architectures whose inductive biases make the acquisition of linguistic structure easier? ¹ Even though modern machine learning algorithms are still far from matching the data efficiency of human children, substantial progress has been made in language modeling. Let us now look at how language modeling architectures has evolved.

1.1.2 Early Language Models

As discussed previously, the earliest language model were n -gram language models. This class of language model is exactly the one Shannon describes: a text is generated by predicting the next symbol from a fixed number of previous symbols, the n -gram context. These symbols can be letters or words, but nowadays they are usually tokens, units that can form a word or a subword. More formally, the probability of picking a token s_i given a sequence of previous tokens $s_{<i} = (s_1, s_2, \dots, s_{i-1})$ is written as $P(s_i \mid s_{<i})$. Using the chain rule, these individual probabilities can be multiplied together to give the probability of the whole text:

$$P(s) = p(s_1) \times p(s_2|s_1) \times p(s_3|s_1, s_2) \times \dots \times p(s_t|s_{<t}) = \prod_{i=1}^t P(s_i \mid s_{<i}). \quad (1.1)$$

¹ For Chomsky, this question is not really interesting, as reproducing linguistic behavior with an artificial architecture is not, by itself, an explanation of human language acquisition:

<https://www.youtube.com/watch?v=PZQ>

Each individual probability can be defined from a score over the vocabulary:

$$p(s_i | s_{<i}) = \frac{\text{score}(s_{<i}, s_i)}{\sum_{v \in \mathcal{V}} \text{score}(s_{<i}, v)}. \quad (1.2)$$

Here, $\text{score}_\theta(x, y)$ measures how strongly the context x is associated with the next token y , and $\mathcal{V}$ is the token vocabulary. The function $\text{score}(x, y)$ can be designed very freely but is often a counting function: how often y appears after the context x in the training corpus.

Using Maximum Likelihood Estimation, the probabilities of the sequence can be estimated by a statistical model parameterized by θ on a large training corpus $\mathcal{C}$ of texts:

$$\theta^* = \arg \max_{\theta} \sum_{s \in \mathcal{C}} \sum_{i=1}^{|s|} \log P_{\theta}(s_i | s_{<i}). \quad (1.3)$$

This learning objective does not really change for generative language models. What changes is how the probabilities are represented and learned.

Larger n -gram sizes usually work better, because they provide more context for predicting the next token. But they also require much more data, because the number of possible token combinations grows exponentially with n . Most long n -grams are hapax, meaning that they appear only once in the training data. In practice, many classical n -gram models were trained up to 5-grams. With web-scale corpora, n -gram language models were estimated on trillions of words², and these models served as a foundation for tasks such as machine translation, speech recognition, and many other text generation tasks.

² <https://research.google/blog/all-our-n-gram-are-belong-to-you/>

But beyond the problem of n -gram size, n -gram language models are also limited by the fact that they do not represent the meaning of the text. They mostly remember local co-occurrences without modelling the meaning of the n -grams. Subsequent research therefore moved toward neural n -gram language models, based on MLPs or CNNs, which learn distributed representations of words during training. Interestingly, the work of Bengio et al. [12], an MLP-based n -gram language model, already shows some of the main engineering ideas for scaling neural language models. By training the MLP to predict the next word, the model learns good representations of the language. The probability of the next token s_i , given the embedding representation $\mathbf{c} \in \mathbb{R}^d$ of the n -gram, is computed using softmax:

$$p_{\theta}(s_i | s_{<i}) = \frac{\exp(\mathbf{c} \cdot \mathbf{W}_{s_i})}{\sum_{v \in \mathcal{V}} \exp(\mathbf{c} \cdot \mathbf{W}_v)} \quad (1.4)$$

where $\mathbf{W} \in \mathbb{R}^{|\mathcal{V}| \times d}$ is the output matrix, which is also called next-token prediction head or **language modeling head (LM Head)**. We will discuss a lot this LM Head, because in small models it can represent a large fraction of parameters and be a bottleneck during generation. Worth noting, the authors pay attention to the engineering

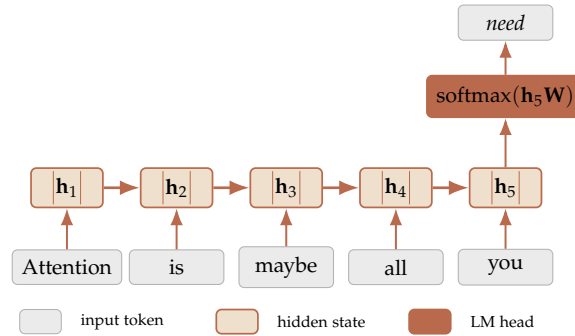
challenges of training such a model. They describe a parallel implementation on multiple CPUs with two strategies: *data-parallel processing*, where different processors work on different training examples, and *parameter-parallel processing*, where the computation over the large output vocabulary is split across processors. These two type of strategies are still used in modern large-scale language model training.

The main limitation of while n -gram language models is that the model size grows linearly with n and that the memory remains short in practice, only a few words. A solution to the context limit of n -gram language models is the Recurrent Neural Network (RNN), and later variants such as LSTM and GRU. Early recurrent networks already proved able to learn linguistic structures over time Elman [35]. Contrary to n -gram models, an RNN does not keep only a fixed window of previous tokens. It reads the sequence token by token and updates a hidden state:

$$\mathbf{h}_{i-1} = \tanh(\mathbf{W}_{hx}\mathbf{x}_{i-1} + \mathbf{W}_{hh}\mathbf{h}_{i-2} + \mathbf{b}_h) \quad (1.5)$$

The hidden state $\mathbf{h}_{i-1}$ acts as a compressed memory of the previous context $s_{<i}$, and is then used to predict the next token s_i :

$$p_\theta(s_i | s_{<i}) = \frac{\exp(\mathbf{h}_{i-1} \cdot \mathbf{W}_{s_i})}{\sum_{v \in \mathcal{V}} \exp(\mathbf{h}_{i-1} \cdot \mathbf{W}_v)} \quad (1.6)$$



In theory, RNNs can model long context because the hidden state compresses the information contained in the whole sequence. This represents a good thing for language modeling. Mikolov et al. [77] showed that RNN-based language model can improve over classical n -gram models by 50% lower perplexity. Connex to language modelling, RNNs have been used a lot in for sequence-to-sequence (Seq2Seq) modeling Cho et al. [21], Kalchbrenner and Blunsom [55], and Sutskever, Vinyals, and Le [101], where an RNN encoder represents the input (text, audio or image) and another RNN decoder which acts as a conditional language model generates the corresponding text. This architecture is useful for tasks such as machine translation, speech recognition, image captioning, etc.

The RNN lacks expressivity as the whole sequence is compressed into a single vector and the transition to the previous state is done by

one single static matrix. Sutskever, Martens, and Hinton [100] tried to enrich the interaction of the current input with the context by making the transformation the previous hidden state input dependant. They achieve this by generating $\mathbf{W}_{hh}$ based on the input. They show that this improves the training of RNNs for character language modeling. Bahdanau, Cho, and Bengio [10] goes even further by avoiding to compress the history: instead of relying only on one final vector, full context is visible and the current input can see the previous hidden states. Let $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_T]$ be the matrix of previous hidden states, where $\mathbf{h}_j$ represents the hidden state at position j . At decoding step i , the current hidden state $\mathbf{h}_i$ is used as a query to determine which previous hidden states are most relevant for the next prediction. The attention weights are computed as:

$$\mathbf{a}_i = \text{softmax} \left(\mathbf{H}^\top \mathbf{h}_i \right), \quad (1.7)$$

where $\mathbf{a}_i \in \mathbb{R}^T$ contains one attention weight for each previous hidden state. The model then constructs the context vector as the weighted sum of the previous hidden states:

$$\mathbf{c}_i = \mathbf{H} \mathbf{a}_i. \quad (1.8)$$

The context $\mathbf{c}_i$ can represent and modulate all the information contained in previous hidden states, instead of relying to a single compressed vector. The context $\mathbf{c}_i$ is used to compute the probability distribution of the next tokens through the LM Head: $\text{softmax}(\mathbf{c}_i \cdot \mathbf{W})$.

While adding attention on top of RNN architecture greatly improve the quality of the generated text in machine translation, the overall architecture is not easy to scale, just as RNNs.

1.1.3 The Transformer, the best candidate for language modeling?

While Banhaudeau et al. improved performance by adding attention on top of RNN layers, Vaswani propose to fully remove the RNN and keep only the attention. This remove the sequential processing imposed by RNN, which make the training parallelisable and hence scalable, since attention is full parallelisable. It also remove all the problems of RNN such as the instabilities etc. but scalability is what made the main success of the Transformer. Figure x presents the transformer architecture for language modeling. The architecture alternates between attention and MLP for N layers and then finally the output, the next token prediction, is computed. Let's detail and discuss these 3 components.

1.1.3.1 Attention

The Transformer haven't introduced attention since we have seen that this has been used by Banhaudeau et al., in an RNN architecture. The

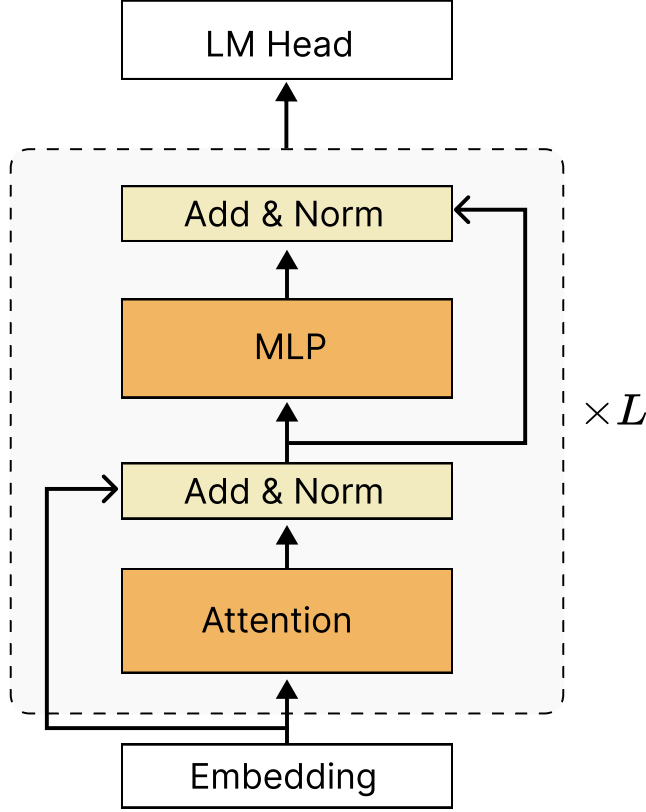


Figure 1.1: Decoder-only Transformer architecture.

innovation of Transformer is to remove the RNN and replace it fully with **Mult-Head Attention (MHA)**. Given $\mathbf{X} \in \mathbb{R}^{T \times d_{\text{model}}}$, the sequence of token embeddings for T positions, MHA projects the sequence into queries, keys and values for H heads:

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{T \times H \times d_{\text{head}}}. \quad (1.9)$$

Equivalently, each head h has its own projections:

$$\mathbf{Q}_h = \mathbf{X} \mathbf{W}_h^Q, \quad \mathbf{K}_h = \mathbf{X} \mathbf{W}_h^K, \quad \mathbf{V}_h = \mathbf{X} \mathbf{W}_h^V, \quad (1.10)$$

where $\mathbf{W}_h^Q \in \mathbb{R}^{d_{\text{model}} \times d_q}$, $\mathbf{W}_h^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ and $\mathbf{W}_h^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ are learned projection matrices for head h . The head output is then computed with scaled dot-product self-attention:

$$\text{head}_h = \text{softmax} \left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top + \mathbf{M}}{\sqrt{d_k}} \right) \mathbf{V}_h. \quad (1.11)$$

For a decoder language model, $\mathbf{M}$ is a causal mask: entries corresponding to future positions are set to $-\infty$, so token i can only attend to tokens $j \leq i$. Multi-head attention runs H such heads in parallel and concatenates their outputs:

$$\text{MHA}(\mathbf{X}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}^O, \quad (1.12)$$

with $\mathbf{W}^O \in \mathbb{R}^{Hd_v \times d_{\text{model}}}$. Each head can therefore learn a different projection of the same context, while the output projection mixes these views back into the residual stream.

In standard MHA, as introduced in the original Transformer paper, the projection matrices $\mathbf{W}_h^Q$, $\mathbf{W}_h^K$ and $\mathbf{W}_h^V$ are usually chosen so that all heads together project back to a space of size d_{model} . However, these projections can become a bottleneck during generation. In a naive implementation, at each decoding step, all previous tokens would be projected again to recompute their keys and values. The **KV-Cache** reduces this computation and speeds up generation by caching the keys and values of previous tokens. However, this cache grows linearly with the sequence length, and it is easy to run out of memory when generating very long texts. To reduce the KV-Cache, we need to reduce the dimensionality of keys and values.

Among the popular approaches to reduce the KV-Cache, **Group Query Attention (GQA)** shares key-value pairs to multiple query heads. Intuitively, the keys and values have fewer heads than the queries:

$$\mathbf{K}, \mathbf{V} \in \mathbb{R}^{T \times h_g \times h_{\text{dim}}}, \quad (1.13)$$

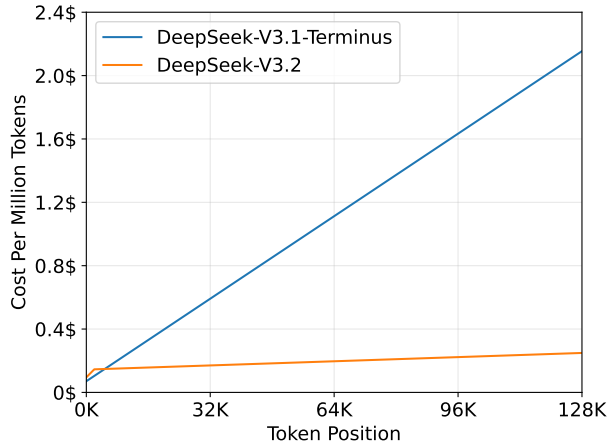
$$\mathbf{Q} \in \mathbb{R}^{T \times h_q \times h_{\text{dim}}}, \quad h_g < h_q. \quad (1.14)$$

The key-value heads are then broadcast to the query heads, and attention is computed as usual:

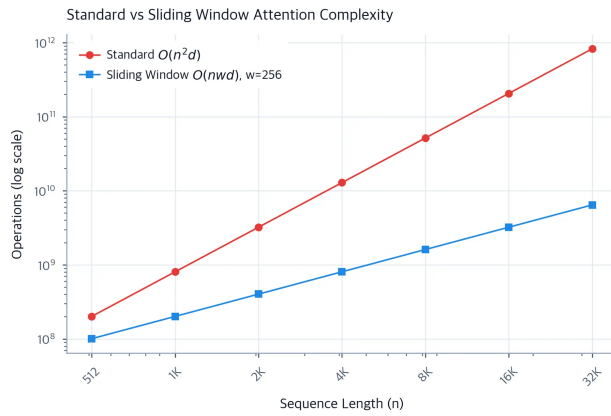
$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^\top) \mathbf{V}. \quad (1.15)$$

Standard MHA corresponds to no sharing, while multi-query attention is the extreme case where all query heads share the same keys and values. GQA is the compromise between the two: it reduces the KV-Cache while keeping several different key-value heads.

While GQA reduces KV-cache memory, it does not reduce the attention computation. **Sparse Attention** reduces this compute by allowing each query to attend to only a subset of previous tokens. DeepSeek Sparse Attention [31] introduces a lightweight *lightning indexer* that cheaply scores all previous tokens using a small number of low-dimensional attention heads. The top-k scoring keys are then selected, and the expensive main attention is computed only over those key-value entries. The main attention therefore changes from $\mathcal{O}(L^2)$ to $\mathcal{O}(Lk)$, where $k \ll L$ is the number of selected tokens. The indexer itself remains $\mathcal{O}(L^2)$, but its constant cost is much smaller because it uses very small dimensions, few heads, and FP8 computation. While many recent models use a derivative of DeepSeek Sparse Attention (cite Qwen, GLM), it is not the first work to introduce sparse attention, which has been widely studied [11, 60, 88, 103]. Sliding-Window Attention is a simple form of sparse attention where each query attends only to a fixed number of nearby tokens instead of the entire preceding



(a) DeepSeek Sparse Attention.



(b) Sliding Window Attention.

Figure 1.2: Sparse-attention decoding costs and computation savings. (a) DeepSeek Sparse Attention (DeepSeek-v3.2) vs DeepSeek with full attention (DeepSeek-v3.1-Terminus) [31]. (b) How Sliding Window Attention reduces computation, from <https://mbrenndoerfer.com/writing/sliding-window-attention>

sequence. For a query at position i and a window size w , attention is restricted to the key-value pairs $(k_{i-w}, v_{i-w}), \dots, (k_i, v_i)$. Each query therefore attends to only w tokens rather than all L previous tokens, reducing the attention complexity from $O(L^2)$ to $O(Lw)$, where typically $w \ll L$. This saves compute and save KV-Cache memory if the implementation discards KV states that fall outside the active window. Sliding-window attention benefits from model depth because information can propagate from one local window to another across successive layers, even though attention is local within each individual layer. However, long-range information is accessed only indirectly through model depth, whereas full attention allows a query to directly attend to any previous token at each layer. For this reason, many LLMs use hybrid architectures that combine local-attention layers with occasional full-attention layers [124–126]. Many of the efficient attentions methods are orthogonal to each other, so they can be combined. For example, it is possible to combine sparse attention with local attention and sliding window attention. Or just as DeepSeek-v4 combine sparse attention with multi-latent attention which is another method for learning compressed kv-cache during pre-training.

1.1.3.2 Multilayer Perceptron

After the attention layer, the contextualized tokens are fed into the Multilayer Perceptron (MLP). While attention is a token-mixing operation, the MLP is a channel-mixing operation, each token is forwarded independently into the MLP. In the original Transformer’s paper, the MLP is defined as:

$$\text{MLP}(x) = f(\mathbf{x}\mathbf{W}_{up})\mathbf{W}_{down}$$

where $f(\cdot)$ is the non-linear activation function, $\mathbf{W}_{up} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{inter}}}$ project the input token $\mathbf{x} \in \mathbb{R}^{d_{\text{model}}}$ in higher dimension, with $d_{\text{model}} \ll d_{\text{inter}}$. And finally, $\mathbf{W}_{down} \in \mathbb{R}^{d_{\text{inter}} \times d_{\text{model}}}$, projects back to the model dimension. Modern LLMs often replace this simple MLP with a gated linear unit variant. Instead of applying only one non-linearity to the expanded representation, the block learns two projections: one projection $\mathbf{V}_{up}$ produces the values, and the other projection $\mathbf{W}_{up}$ gates them. Omitting bias terms, these alternatives can be written as:

$$\text{MLP}_{\text{GLU}}(\mathbf{x}) = (\sigma(\mathbf{x}\mathbf{W}_{up}) \odot \mathbf{x}\mathbf{V}_{up}) \mathbf{W}_{down}, \quad (1.16)$$

$$\text{MLP}_{\text{GEGLU}}(\mathbf{x}) = (\text{GELU}(\mathbf{x}\mathbf{W}_{up}) \odot \mathbf{x}\mathbf{V}_{up}) \mathbf{W}_{down}, \quad (1.17)$$

$$\text{MLP}_{\text{Bilinear}}(\mathbf{x}) = (\mathbf{x}\mathbf{W}_{up} \odot \mathbf{x}\mathbf{V}_{up}) \mathbf{W}_{down}, \quad (1.18)$$

$$\text{MLP}_{\text{SiLU}}(\mathbf{x}) = (\text{SiLU}(\mathbf{x}\mathbf{W}_{up}) \odot \mathbf{x}\mathbf{V}_{up}) \mathbf{W}_{down}, \quad (1.19)$$

$$\text{MLP}_{\text{ReLU}}(\mathbf{x}) = (\text{ReLU}(\mathbf{x}\mathbf{W}_{up}) \odot \mathbf{x}\mathbf{V}_{up}) \mathbf{W}_{down}. \quad (1.20)$$

Here $\odot$ denotes element-wise multiplication. 1.1 gives the performance of each activation on SuperGLUE benchmark. We can see that SiLU

Average SuperGLUE score	
MLP _{GLU}	73.95
MLP _{GEGLU}	73.96
MLP _{Bilinear}	73.81
MLP _{SwiGLU}	74.56
MLP _{ReGLU}	73.66

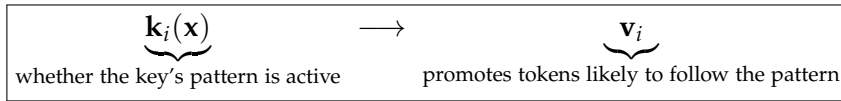
Table 1.1: Average SuperGLUE score for gated MLP variants.

activation gives better results, this explains why many LLMs use it as activations: Mistral-v0.1, Granite, etc. It is difficult to explain why gated MLP with work better with some activations than other. Citing Shazeer [95], the author of gated MLPs:

“We offer no explanation as to why these architectures seem to work; we attribute their success, as all else, to divine benevolence.”

— Noam Shazeer, *GLU Variants Improve Transformer* (2020)

In modern LLM architectures, the MLP parameters can occupy over to 90% of the total model parameters. It has been shown that MLP layers play an important role in storing knowledge acquired during web-scale pretraining. Geva et al. [43] interpret the MLP as a key–value memory: each column of $\mathbf{W}_{\text{up}}$ in the MLP represents a key $\mathbf{k}_i$ that detects linguistic patterns seen during training, while each row of $\mathbf{W}_{\text{down}}$ represents the corresponding value $\mathbf{v}_i$, which promotes the tokens likely to follow those patterns. The key $\mathbf{k}_i$ can be strongly or weakly activated depending on the input x . When $\mathbf{k}_i$ is strongly activated, its corresponding value $\mathbf{v}_i$ is assigned a large weight and therefore contributes more strongly to the MLP output. This mechanism can be schematized as follows:



The MLP does not only store lexical knowledge. At some point, this lexical knowledge coincides with factual knowledge. When a key detecting the prefix *The capital of France is* is strongly associated with a value representing *Paris*, this lexical key–value memory turns out to also encode world knowledge. Hence, more broadly, it is widely believed that MLPs play a central role in storing the knowledge acquired during pretraining Dai et al. [30] and Geva et al. [43]. This view is also reflected in many knowledge-editing methods, which often directly target MLP weights Fang et al. [36] and Meng et al. [74, 75].

Value	Top prediction	Precision@50	Matching trigger example
$\mathbf{v}_{222}^{15}$	<i>each</i>	68%	<i>But when bees and wasps resemble each</i>
$\mathbf{v}_{752}^{16}$	<i>played</i>	16%	<i>Her first role was in Vijay Lalwani's psychological thriller Karthik Calling Karthik, where Padukone was cast as the supportive girlfriend of a depressed man (played</i>
$\mathbf{v}_{2601}^{13}$	<i>extratropical</i>	4%	<i>Most of the winter precipitation is the result of synoptic-scale, low-pressure weather systems (large-scale storms such as extratropical</i>
$\mathbf{v}_{881}^{15}$	<i>part</i>	92%	<i>Comet served only briefly with the fleet, owing in large part</i>
$\mathbf{v}_{2070}^{16}$	<i>line</i>	84%	<i>Sailing from Lorient in October 1805 with one ship of the line</i>
$\mathbf{v}_{3186}^{12}$	<i>jail</i>	4%	<i>On May 11, 2011, four days after scoring 6 touchdowns for the Slaughter, Grady was sentenced to twenty days in jail</i>

Table 1.2: Examples of MLP key–value memory pairs identified by Geva et al. [43]. For each value vector $\mathbf{v}_i^\ell$, *Top prediction* is the vocabulary token receiving the highest score when the value is projected through the model’s output embedding. *Precision@50* is the percentage of the 50 training prefixes that most strongly activate the paired key $\mathbf{k}_i^\ell$ whose ground-truth next token equals that prediction. The final column presents one such matching prefix, with its ground-truth next token shown in blue.

1.1.3.3 Language Modeling Head

Well, the tokens are contextualized thanks to the multi-head attention, and each token retrieves the relevant knowledge encoded within the MLP. This process is repeated for L layers and at the end produces a context vector $\mathbf{c}$ that is ready for next token prediction using the language modeling head (LM Head or output embedding). The LM Head is another important component in large scale Transformer LLM. First, the token vocabulary becomes quite large due to the diversity of tokens that can be inferred from a web scale data. Hence, the LM Head weight can occupy a huge amount of parameters in the LLM, particularly for small LLMs. Figure 1.3 shows the memory occupied by the input/output embedding parameters for different model sizes, including dense and quantized ones. We can see that the embedding

and LM Head parameters represent a larger fraction of the model when the Transformer backbone is small³. This is mainly caused by the

³ Quantization is usually not applied to the embedding weights.

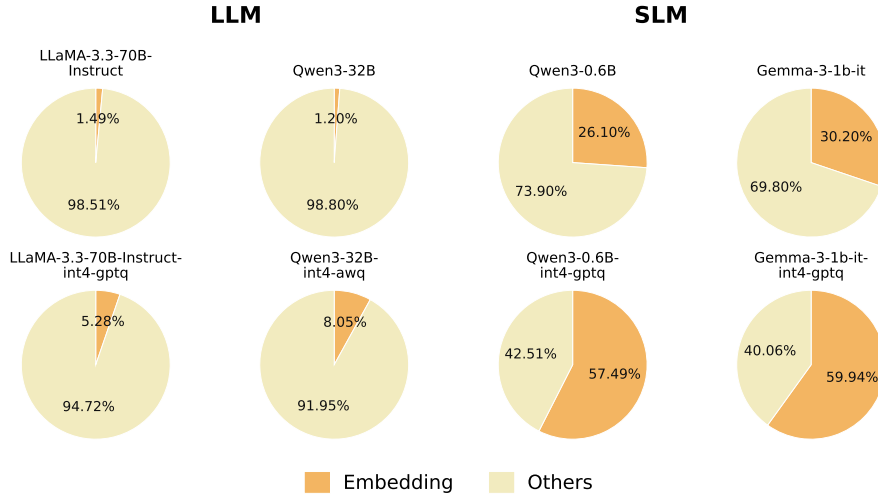


Figure 1.3: Proportion of the size in MB of embedding parameters across different model sizes. Data taken from <https://blog.squeezebits.com/vocabulary-trimming-methods>.

large vocabulary of the models. A common solution to reduce the size by half is weight tying, which shares the input and output embedding. Almost all small LLMs use weight tying, which haven't proven to hurt parameters, while still reducing a lot the model's parameters.

Another approach to reducing the number of parameters occupied by the LM head is to factorize its weight matrix into two low-rank matrices. This strategy was notably adopted in ALBERT [64], where the embedding dimension was reduced from 768 to 128 without sacrificing downstream performance. More precisely, given the input $\mathbf{x} \in \mathbb{R}^d$ and an LM-head weight matrix $\mathbf{W} \in \mathbb{R}^{d \times V}$, the factorized projection is

$$\mathbf{xW} \longrightarrow (\mathbf{xW}_1)\mathbf{W}_2, \quad (1.21)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times r}$, $\mathbf{W}_2 \in \mathbb{R}^{r \times V}$, and $r \ll H$. This reduces the LM-head parameter count from dV to $dr + rV$.

However, some works have identified what is known as the *softmax bottleneck* [yang2018breakingsoftmaxbottleneck]: a low-dimensional LM head can only produce a limited variety of output distributions. Therefore, reducing its rank too aggressively can limit the model's ability to predict complex token distributions. Godey, Clergerie, and Sagot [45] studied this by compressing the LM Head rank dimension at different sizes and found that smaller rank hurts performance and limits the expressiveness of the probability distribution as shown in Figure 1.4. More importantly, this also hurts the learning signal during training Godey and Artzi [44]. There are many solutions to avoid the *softmax bottleneck*. Since this phenomenon arises mainly when $r \ll V$, an obvious solution is to augment r or V the vocabulary size.

Reducing the vocabulary size can be achieved by using a tokenizer with few tokens or falling back to a character or bytes tokenizers. It is also technically possible after training to remove tokens in the input/embedding embedding. But this is risky as it can affect critical tokens. This discussion is related to the research we will present later, which proposes to reduce the rank of the LM Head instead of pruning tokens in multilingual transformer models.

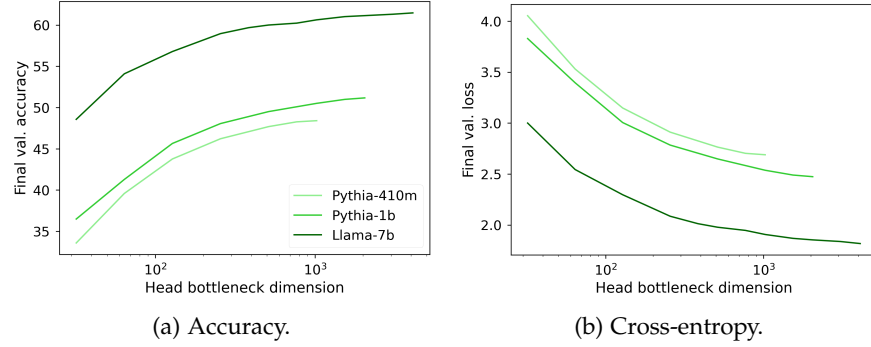


Figure 1.4: Effect of reducing the LM Head rank on accuracy and cross-entropy. Figures from Godey, Clergerie, and Sagot [45].

1.1.4 A Transformer is never enough large

What made the Transformer architecture particularly successful in language modeling is its ability to scale both the number of parameters and the amount of training data. Since Transformer training can be efficiently parallelized across multiple GPUs, increasingly large models can be trained on increasingly large datasets, leading to improved performance and the emergence of new capabilities. However, scaling cannot be done blindly: given the high cost of pretraining, the allocation between model size and training data must be estimated before the final training run. For a fixed compute budget, how large should the model be, and on how many tokens should it be trained?

Denoting the number of parameters by N , the number of training tokens by D , and the training compute by C , the compute cost of a dense Transformer can be approximated as $C \approx 6ND$. So given a training budget C fixed (how many training flops is available for training), how much the model size should take in this compute and how much the training tokens should take in this budget. Both Kaplan [57] and Chinchilla [51] scaling find that the compute-optimal model size and dataset size follow power laws in the compute budget: $N_{\text{opt}} \propto C^\alpha$ and $D_{\text{opt}} \propto C^\beta$. **Kaplan scaling laws** estimate that the compute-optimal allocation with $N_{\text{opt}} \propto C^{0.73}$ and $D_{\text{opt}} \propto C^{0.27}$. Thus, model size should increase much faster than training data, which favors very large models trained on comparatively few tokens. On the

other side, the **Chinchilla scaling laws** later find that model size and training data should instead scale approximately equally: $N_{\text{opt}} \propto C^{0.5}$ and $D_{\text{opt}} \propto C^{0.5}$. Since Kaplan and Chinchilla scaling laws, lot of types scaling laws are proposed for training LLMs. Often it is about adding new scaling axis such as sparsity and granularity in the case of sparse Transformer model. It can also be about accounting a constrain such as an inference budget.

Beyond guiding compute allocation, scaling laws also reveal how model capabilities evolve with scale. While performance often improves predictably as training compute increases, some abilities appear only after the model reaches a certain scale. The model may remain close to random performance across several scales before suddenly crossing a plateau and acquiring a new ability. Wei et al. [110] identify several such *emergent abilities*.

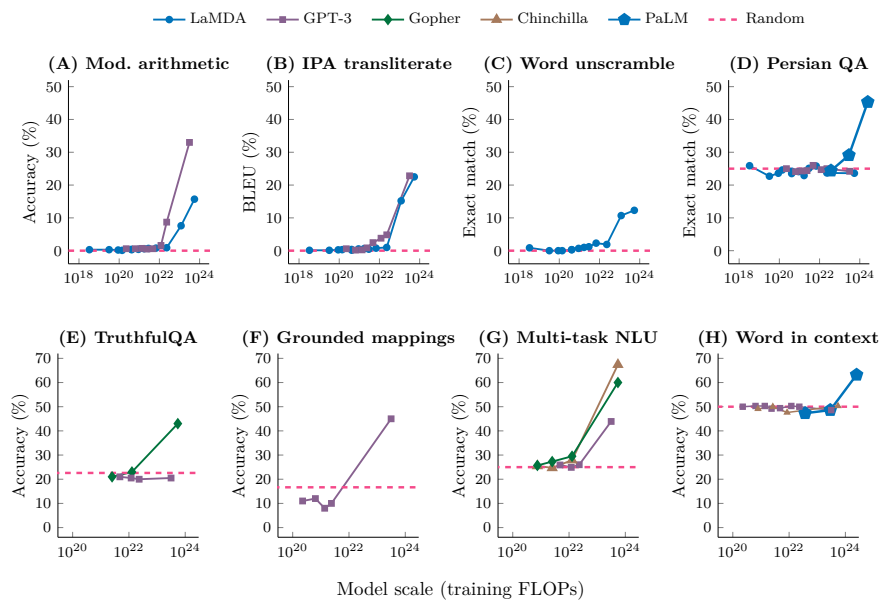


Figure 1.5: Examples of emergent abilities in large language models. Figure from Wei et al. [110]. A capacity is emergent when the score surpasses the random baseline while it wasn't at smaller scale.

Then a question arises. Is there a point beyond which increasing the number of parameters no longer produces meaningful improvements? Testing this hypothesis directly is difficult, since training models at the largest scales is extremely costly. Classical scaling laws predict *diminishing returns*: test loss continues to decrease with scale, but each additional increase in model size produces a smaller improvement [51, 57]. However, whether this implies diminishing returns in model capabilities depends on how performance is measured. Sinha et al. [97] argue that short-task benchmarks may create an illusion of saturation. Even small improvements in single-step accuracy can compound into large improvements in the length of tasks that a model can reliably

execute. Therefore, diminishing returns on loss or short-task accuracy do not necessarily imply diminishing returns in practical capabilities.

1.2 HOW TO MAKE AN LLM MORE EFFICIENT

We've discussed Transformer and how scaling them yields strong improvement in natural language tasks. However, scaling a Transformer-based language models is costly in training compute but also in inference budget. We present here common approaches to make them efficient to scale and This part is heavily inspired by the blog-post we wrote in collaboration with the Datacraft team in 2024: <https://medium.com/@yayasy-nlp/a-comprehensive-overview-of-large-language-models-compression-methods-8b59ddoff4f8> An efficient model minimizes the computations, the GPU memory usage, the weights movement (Bandwidth).

1.2.1 *Model compression*

1.2.1.1 *Llama small model or how to hack the Chinchilla scaling laws*

1.2.1.2 *Quantization*

1.2.1.3 *Weights Sparsity*

UNSTRUCTURED SPARSITY

SEMI-STRUCTURED SPARSITY

1.2.1.4 *Structured Pruning*

1.2.1.5 *The best training objective for compression*

1.2.2 *Activation Sparsity*

1. Scaling model parameters without increasing computation

1.2.2.1 *Mixture of Experts*

1. Mixture of Experts as block activation sparsity
2. Formulation of MoE Block (routing, gating, experts)
3. Limitations of MoE:
 - a) Load balancing
 - b) Dead experts
 - c) Variability in expert choice during inference (expert replay in RL training)
 - d) TopK: non differentiability, non adaptivity

1.2.2.2 *Embedding Memory Layers*

1.2.2.3 *Product Key Memory Layers*

1. Motivations: extreme granularity, offloading
2. Formulation of Sparse Memory Layer (routing, gating, memory)
3. PKM, Scaling PKM, Mixture of million Experts
4. Don't forget the paper by Sakana AI

1.2.2.4 *Matryoshka Representation Learning*

1. MRL as a block activation sparsity
2. Formulation of MRL training
3. MRL as a block sparsity
4. Limitations of MRL:
 - a) Non differentiability of the indexing function
 - b) Non adaptivity

1.2.2.5 *Post-training Activation Sparsity*

1. Teal
2. Cats
3. Relufication
4. Moefication
5. LoSparse
6. Learning to be efficient

1.2.3 *Speculative Decoding and Diffusion LLMs: Trading Compute for Memory*

1. New Gemma paper: profiling of AR vs Diffusion
2. why prefer speculative decoding over diffusion LLMs: <https://x.com/barrowjoseph/status/2086933>

Part II

COMPRESSION OF TRANSFORMER MODELS

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2.1 INTRODUCTION

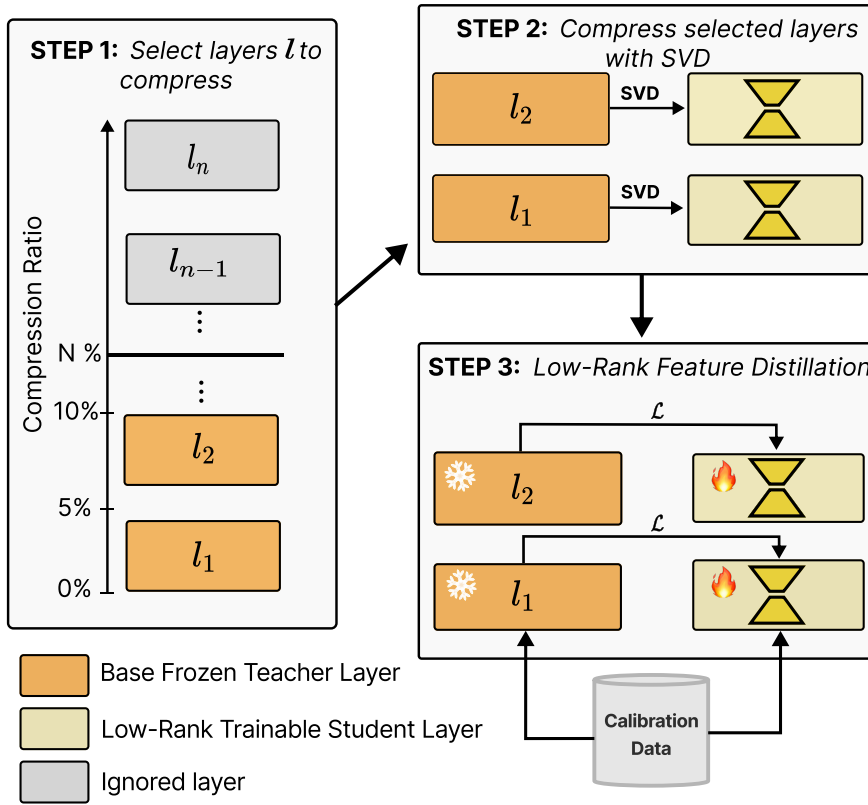


Figure 2.1: **Lillama approach**: **STEP 1** selects layers to compress for a target compression ratio (e.g., $N\%$) using various strategies (see Section 2.6). **STEP 2** compresses and initializes the chosen parameters via SVD. **STEP 3** distills the low-rank weights with a small calibration dataset.

As discussed earlier, compressing Large Language Models is a crucial challenge that can be achieved with several complementary approaches: quantization, pruning, compression, and distillation. In this section, we introduce a new one-shot compression method that locally distills low-rank weights and satisfies the three objectives:

1. The compression algorithm must be compute-efficient, e.g., it should require minimal computational resources and run in a reasonable amount of time. For instance, the iterative magnitude pruning algorithm as done in Frankle and Carbin [37], is too costly to be applied to large-scale LLMs.

Method	No Custom Kernel	No Ampere-only	Memory Gain	Is One-Shot	Calib. Data
SparseGPT [38]	✗	✗	✗	✓	-
Wanda [99]	✗	✗	✗	✓	-
Teal [71]	✗	✗	✗	✓	-
ShearedLLaMA [116]	✓	✓	✓	✗	52B
Minitron [98]	✓	✓	✓	✗	94B
Lillama	✓	✓	✓	✓	0.013B

Table 2.1: Positioning of our approach within the compression literature. **No Custom Kernel** indicates that a custom GPU kernel is not required to observe speedup and **No Ampere-only** is whether the method requires Ampere GPU to achieve speedup. **Calib. Data** is the total number of tokens in billions used for compression.

2. It must achieve fast convergence using as little training data as possible; this objective is often an issue with many model pruning methods that often require significant retraining after pruning to recover the lost performance.
3. The method should not cause a significant drop in performance; ideally, all the capabilities of the original LLM should remain intact.

To address the first constraint, we compress the model with a local distillation objective, as opposed to a global objective that is more costly. We further reduce the cost by compressing and distilling only a subset of all layers. About the second constraint, we initialize the compressed layers with Singular Value Decomposition (SVD), which enables reducing the required number of gradient steps as well as the calibration dataset size. We further improve convergence by combining Teacher and Student activations with a joint distillation loss. We experimentally show the robustness of our approach by compressing several state-of-the-art small and large Transformers, Mixture-of-Experts, and Mamba LLMs. Figure 2.1 gives an intuitive overview of the proposed method, which is detailed in Section 2.6 and validated in Section 2.7.

2.2 RELATED WORKS

We discuss next previous compression methods, highlight their limitations, and explain how our approach addresses them. Table 2.1 positions our approach relative to recent compression methods.

2.2.1 *Pruning*

Pruning methods remove unimportant weights in the pre-trained model LeCun, Denker, and Solla [65], Han et al. [47]. Pruning can be applied to models in different ways and there are three pruning families:

2.2.1.1 *Unstructured Pruning*

In unstructured pruning, individual parameters are removed, which results in a model with sparse weights. As is, this approach does not provide memory savings or speedup without dedicated algorithmic tooling. Since zeros are placed in pre-trained weights in an unstructured and unpredictable way, it is difficult to exploit this on GPU tensor cores. However, when many parameters are pruned, such that there are many zeros in the weight matrix, tensor multiplication can be accelerated with dedicated GPU kernels. Similarly, some tensor storage formats can provide substantial memory gains for an unstructuredly pruned model.

2.2.1.2 *Semi-Structured Pruning*

Contrary to unstructured pruning, semi-structured pruning enforces fixed local sparsity patterns (e.g., N:M), as in Sun et al. [99], Frantar and Alistarh [38]. These approaches often do not lead to memory gains but can speed up processing on kernels optimized for such sparse patterns. On the other hand, our method, which directly shrinks matrices, saves memory across all hardware and leads to speedup, as fewer computations are performed.

2.2.1.3 *Structured Pruning*

Structured pruning removes entire groups of parameters, which results in a smaller and faster model. Ma, Fang, and Wang [73] propose a gradient-based criterion to eliminate substructures in LLMs, while Xia et al. [116] use a joint loss combining a pruning mask loss with the language modeling loss. However, optimizing these criteria can be computationally intensive. For example, the pruning step of Sheared-LLaMA Xia et al. [115] is 5x expensive compared to standard LM training, according to the authors. In contrast, thanks to the local gradient updates, our approach is computationally efficient, allowing us to compress a 47B model within minutes on a single A100 GPU.

2.2.2 *Low-Rank Decomposition*

Low-Rank Decomposition compresses a model by approximating its pre-trained matrices with lower-dimension ones. Based on the observa-

tion by Li et al. [66] that Neural Networks have lower intrinsic dimensions, Aghajanyan, Zettlemoyer, and Gupta [3] show that Transformer language models also require lower intrinsic dimensions depending on the language task. However, Yu and Wu [119], Chen et al. [20] show that the weights in some pretrained Transformer models are full-rank compared to their activations and propose an activation-aware compression method.

Our work is related to these approaches, as we also decompose the matrices of the models. However, we apply a lightweight layer-wise feature distillation objective to better recover from compression and propose different distillation strategies to accelerate convergence and improve performance.

2.2.3 Distillation

In Knowledge Distillation (KD), a small student model learns to reproduce the outputs of a larger pre-trained model. Standard KD is data-intensive when the student model’s parameters are randomly initialized. Many recent open-source LLMs such as Gemma-4 or Llama-4, are distilled from a larger model using trillion of tokens, which is as costly as a standard pretraining stage.

Another approach is layer-wise distillation, which can be more suitable in some cases, as intermediate layers may contain certain information, such as speaker identity or phonemes in pretrained speech models Baevski et al. [8], Pasad, Chou, and Livescu [82], Chang, Yang, and Lee [18], Chang et al. [19]. For example, TinyBERT Jiao et al. [54] initializes the student model by removing layers from the teacher and then distills the student on task-specific knowledge, combining layer-wise and global distillation losses, which took 3.5 days to train. This is costly for LLMs. Instead of removing layers, we leverage the observation that layer activations are low-rank and approximate them with fewer parameters. Additionally, we compress the models to remain generalist rather than task-specific.

2.3 MOTIVATION: ACTIVATIONS ARE MORE LOW-RANK THAN WEIGHTS

In order to motivate the compression of the the models weights with SVD, the weights should have a low low-rank structure. However, previous studies Chen et al. [20], Yu and Wu [119] have shown that the weights of pretrained Transformer models are often full-rank, while their activations are more low-rank. We empirically confirm this observation on three recent models: Falcon-Mamba, Mistral-v0.1 7B, and Phi-2 3B. There are different measure for rank. In this study, we

use a simpler one: stable rank (*srank*) as a proxy measure for rank. The stable rank of a matrix is defined as the following ratio:

$$\text{srank}(X) = \frac{\sum_{i=1}^r \sigma_i^2}{\sigma_1^2} \quad (2.1)$$

with $X \in \mathbb{R}^{d_1 \times d_2}$ being the input matrix (activations or weights), $r = \min(d_1, d_2)$, and $\sigma_1 \geq \dots \geq \sigma_r$ the singular values of X .

Figure 2.2 shows the stable rank of various weight matrices and layer activations for three models: Falcon-Mamba, Mistral-v0.1 7B, and Phi-2 3B. For all models, activations are significantly lower-rank than weight matrices, which confirms previously published results by Yu and Wu [119].

This does not only motivate the use of activations for compression, but also explains why applying SVD to weights alone does not work well as we will show in the experiments. How to explain the fact that weights are full-rank while activations are low-rank? This is an open question. We know that modern LLMs are pre-trained on large and diverse datasets from the web that represent lot of knowledge. Hence the weights of the model need to be full-rank to capture all this knowledge. However, for a given input of a specific task, only a subset of this knowledge is needed, which may explain the low-rank structure of activations.

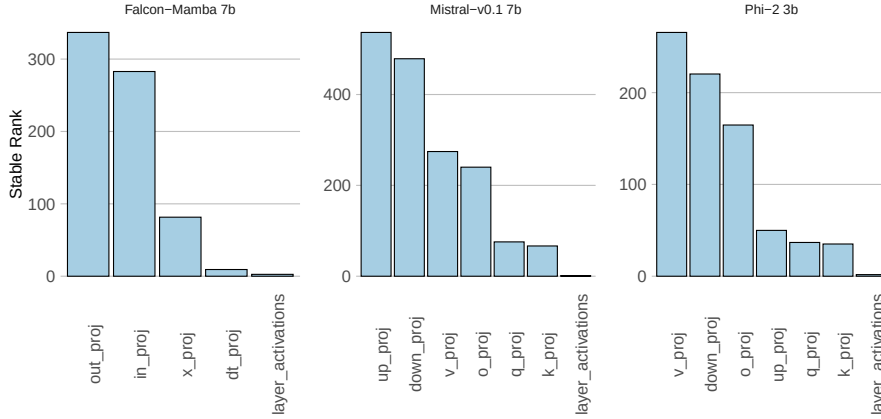


Figure 2.2: **Activations are low-rank.** Comparison of the stable rank of weights (*_proj) and layer activations (layer_activations). Each bar is the average stable rank across all layers.

2.4 BACKGROUND: LOW-RANK APPROXIMATION OF TRANSFORMER MODELS

Before presenting our approach, we first introduce the two methods for low-rank approximation of Transformer models: low-rank approximation from weights and from features.

2.4.1 Low-Rank Approximation from Weights

Given $W \in \mathbb{R}^{d_1 \times d_2}$, a pretrained weight matrix, and an example $x \in \mathbb{R}^{d_2}$, the activations $y \in \mathbb{R}^{d_1}$ are computed as $y = Wx$. It is possible to reduce the computations in this operation by using a low-rank approximation of W : $y \simeq \Delta W x = ABx$ with $\Delta W = AB$ the low-rank decomposition of W , composed of $A \in \mathbb{R}^{d_1 \times r}$ and $B \in \mathbb{R}^{r \times d_2}$. When the rank r is small enough, the number of parameters $r(d_1 + d_2)$ in the low-rank equation is smaller than $d_1 d_2$ in the full-rank equation. Estimating the low-rank matrix $\Delta W = AB$ can be formulated as a minimization problem to find the low-rank ΔW that best approximates W :

$$\widehat{\Delta W} = \underset{\Delta W}{\operatorname{argmin}} \|W - \Delta W\|_F \quad (2.2)$$

It is well-known that this minimization problem can be approached using SVD, a low-rank approximation method that offers the optimal r -rank approximation of a given matrix with regard to the Frobenius norm $\|\cdot\|_F$. SVD approximates a matrix W into three matrices: $W = USV^T$, where $S \in \mathbb{R}^{d_1 \times d_2}$ is a diagonal matrix containing the singular values of W sorted in descending order, $U \in \mathbb{R}^{d_1 \times d_1}$ and $V \in \mathbb{R}^{d_2 \times d_2}$ are orthonormal matrices. The optimal low-rank approximation of $W \in \mathbb{R}^{d_1 \times d_2}$ can be obtained by keeping the first r singular values, with $r < \min(d_1, d_2)$:

$$\widehat{\Delta W} = (U_{:,r} S_{:,r,r}) V_{r,:} = AB \quad (2.3)$$

with $A = U_{:,r} S_{:,r,r}$ and $B = V_{r,:}$ and the notation $_{a:b}$ refers to the slicing operation.

2.4.2 Low-Rank Approximation from Feature

As discussed previously, many studies Chen et al. [20], Yu and Wu [119] have shown that the activations (i.e., features) of pretrained transformers are more low-rank than the weights. Recently, Liu et al. [71] show that transformer activations can be sparsified up to 60%. Hence the motivation to use the activations of the linear layers to estimate the low-rank matrices.

Let $\mathcal{D} = \{x_i \in \mathbb{R}^{d_2}\}_{1 \leq i \leq N}$ be a calibration dataset, and $W \in \mathbb{R}^{d_1 \times d_2}$ the parameters of a linear layer. Finding the low-rank matrix ΔW that best reproduces the activations $\{y = Wx\} \quad \forall x \in \mathcal{D}$ involves solving:

$$\widehat{\Delta W} = \underset{\Delta W}{\operatorname{argmin}} \frac{1}{N} \sum_{x \in \mathcal{D}} \|Wx - \Delta Wx\|_F \quad (2.4)$$

An analytic solution to this minimization problem is the eigendecomposition of the covariance matrix of the activations. We first collect

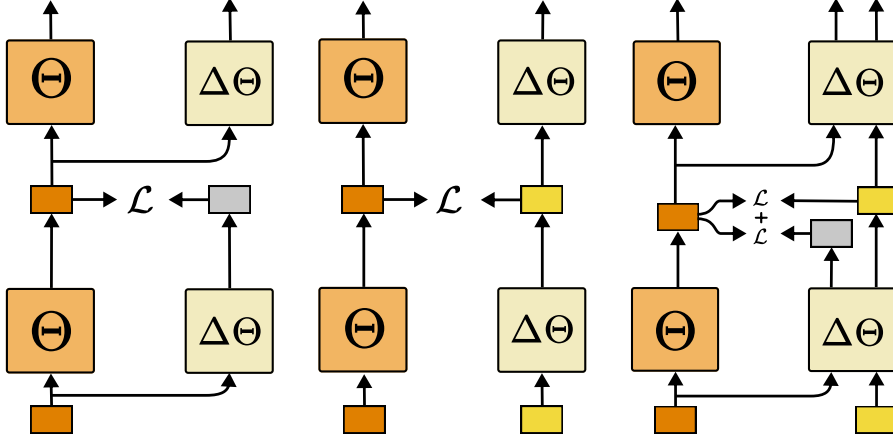


Figure 2.3: **Accelerating convergence through a joint loss.** We compare three distillation strategies: (a) Teacher, (b) Student, and (c) Teacher+Student.

all activations $\mathcal{Y} = \{y = Wx\}_{\forall x \in \mathcal{D}}$, then the covariance matrix of the activations can be estimated as $\Sigma = \mathbb{E}_{y \in \mathcal{Y}} [yy^T] - \mathbb{E}[y]\mathbb{E}[y]^T$ with $\Sigma \in \mathbb{R}^{d_1 \times d_1}$. Since Σ is a diagonalizable matrix, we can apply its eigendecomposition: $\Sigma = USU^T$, where $U \in \mathbb{R}^{d_1 \times d_1}$ contains the eigenvectors of Σ and $S \in \mathbb{R}^{d_1 \times d_1}$ is the diagonal matrix containing the eigenvalues sorted in decreasing order. As for the SVD equation above, we can keep the eigenvectors corresponding to the r largest eigenvalues, which gives us $A = U_{:,r}$ and $B = U_{:,r}^T W$.

Compared to the low-rank matrices in the SVD equation above, A and B weights here are learned to reproduce the activations of the base weight W , thanks to the minimization objective above. In the next section, we highlight the limitations of this approach and introduce our proposed method.

2.5 PROPOSED APPROACH

We identify and handle next three potential limitations of the approach presented in Section 2.4.2.

2.5.1 Non-linear Feature Approximation.

The previous analytical solution above is limited to activations from linear layers. To generalize to non-linear modules, we first observe that this objective can be seen as a feature distillation objective that may be optimized numerically by gradient descent rather than analytically by eigendecomposition. Given the input batch $X \in \mathbb{R}^{d \times b}$ of b examples, we denote the output activations of the i^{th} Teacher module $\mathcal{T}^{(i)}$ as

$$Y^{(i)} = \mathcal{T}^{(i)}(X; \Theta^{(i)}) \quad (2.5)$$

where $\Theta^{(i)}$ are the original pretrained matrices of the i^{th} Teacher, and $Y^{(i)} \in \mathbb{R}^{d \times b}$ its output activations. Similarly, we note the output activations of the i^{th} Student module $\mathcal{S}^{(i)}$ as:

$$\hat{Y}^{(i)} = \mathcal{S}^{(i)}(X; \Delta\Theta^{(i)}) \quad (2.6)$$

where the Student module is parametrized with the low-rank matrices $\Delta\Theta^{(i)}$ and $\hat{Y}^{(i)} \in \mathbb{R}^{d \times b}$ are the output activations. We can express the distillation objective of the i^{th} Student module as:

$$\widehat{\Delta\Theta}^{(i)} = \underset{\Delta\Theta^{(i)}}{\operatorname{argmin}} \mathcal{L}^{(i)}(Y^{(i)}, \hat{Y}^{(i)}) \quad (2.7)$$

where $\widehat{\Delta\Theta}^{(i)}$ are the estimated low-rank matrices of the i^{th} Student module and $\mathcal{L}^{(i)}(Y^{(i)}, \hat{Y}^{(i)})$ is the loss measuring the distance between the activations of the Teacher and the Student. We opted for the same loss as Chang, Yang, and Lee [18] because we observed that the loss yields instabilities:

$$\begin{aligned} \mathcal{L}^{(i)} &= \mathcal{L}_{\ell_1}^{(i)} + \mathcal{L}_{\cos}^{(i)} \\ &= \sum_{t=1}^b \left[\frac{1}{D} \left\| Y_t^{(i)} - \hat{Y}_t^{(i)} \right\|_1 - \log \sigma \left(\cos \left(Y_t^{(i)}, \hat{Y}_t^{(i)} \right) \right) \right] \end{aligned} \quad (2.8)$$

where D is the hidden vectors dimension, σ is the sigmoid activation and $\cos(\cdot, \cdot)$ is the cosine similarity. We propose to approximate the low-rank weights of the Students through Gradient Descent. In the linear case, this should converge towards the eigendecomposition solution described in Section 2.4.2. However, as we show later, we extend the distillation process to non-linear modules, which requires numerical optimization. We also show that initializing the Student's low-rank parameters with SVD, rather than randomly, improves convergence.

2.5.2 Beyond Teacher-only Activations.

The second potential limitation of the approach described in Section 2.4.2 and equations above is that the Student and the Teacher modules take the same input X , which is the output of the previous layer of the Teacher. This means that the Student is trained from the activations of the Teacher module only, which are not available at inference time. Yu and Wu [119] use this approach to approximate the Linear Layers of Transformer models, referring to it as Atomic Feature Mimicking. Let $\hat{Y}_T^{(i)} = \mathcal{S}^{(i)}(Y_T^{(i-1)}; \Delta\Theta^{(i)})$ be the output activations of the Student module when fed $Y_T^{(i-1)}$, the output activations of the Teacher at the previous layer $i - 1$. The loss can then be written as:

$$\mathcal{L}_T^{(i)} = \mathcal{L}^{(i)}(Y_T^{(i)}, \hat{Y}_T^{(i)}) \quad (2.9)$$

where $Y_T^{(i)}$ are the gold activations of the current Teacher. This approach can lead to a fast convergence as the Student modules learn

from gold and high-quality activations produced by the Teacher modules. However, it can cause a performance drop during inference as the Teacher activations are not available. An alternative approach is to have the Student module take as input the activations of previous Student modules rather than those of Teacher modules:

$$\mathcal{L}_S^{(i)} = \mathcal{L}^{(i)}(Y_T^{(i)}, \hat{Y}_S^{(i)}) \quad (2.10)$$

where $\hat{Y}_S^{(i)} = \mathcal{S}^{(i)}(\hat{Y}_S^{(i-1)}; \Delta\Theta^{(i)})$ is the output activations of the current Student module when taking as input the output activations $\hat{Y}_S^{(i-1)}$ of the Student module at the previous layer, and $Y_T^{(i)}$ are the gold activations of the corresponding Teacher module. However, although we found this loss performs generally better than the previous one, it leads to a slow convergence due to the errors in the activations of the Student modules. To address both issues, we propose to combine the two losses:

$$\mathcal{L}_{T+S}^{(i)} = \mathcal{L}_T^{(i)} + \mathcal{L}_S^{(i)} \quad (2.11)$$

This distillation approach is illustrated in Figure 2c. We don't introduce any hyperparameter for controlling the importance of one loss over another in the joint loss. We leave this for future work.

2.5.3 Module-wise Distillation.

Finally, we observe that the functions can be any other module in the neural network than linear layers: any sub-part of the model that outputs low-rank activations can be compressed with our method. In our experiments, we distill at the Transformer layer level and leave comparisons with other distillation levels for future work. We also experiment with other models than pretrained Transformer-based language models.

2.6 FAST AND MEMORY-EFFICIENT COMPRESSION

Computing the optimal compression rank of each matrix individually may be costly: while search algorithms can be a solution for smaller models, they can be difficult to scale to LLMs with billions of parameters. Assuming a target compression rate N , we consider three simple strategies: **uniform**, **top**, and **bottom**-first compression. Figure 2.4 summarizes the strengths and weaknesses of each strategy.

2.6.1 Uniform Weights Compression

This approach removes $N\%$ parameters from each weight matrix in the model. This approach doesn't make any assumption about the importance of different layers but is also costly because it requires

load only a subset of the model in GPU memory and forward through only a subset of the first compressed layers, which is not the case for the previous two approaches. As we will show it, this approach can compress big models that could not fit into GPU memory. The Algorithm 1 describes the bottom-first compression approach.

Algorithm 1: Bottom Layers First Compression Algorithm

Input: $\mathcal{M}$ is the base model; S is the target size of the compressed model; k is the minimum possible rank to set; m the increment when generating ranks
Result: Low-rank compressed $\mathcal{M}$

```

1  $\mathcal{R} \leftarrow \emptyset$ ;
2 for each weight matrix  $W \in \mathbb{R}^{d_1 \times d_2}$  in  $\mathcal{M}$  do
3    $b \leftarrow \min(d_1, d_2)$ ;
4   for  $r = k$ ;  $r \leq b$ ;  $r = r + m$  do
5     if  $r \times (d_1 + d_2) < d_1 \times d_2$  then
6        $\mathcal{R} \leftarrow \mathcal{R} \cup \{(r, W)\}$ ;
7 sort  $\mathcal{R}$  by layer index in increasing order, then by rank in decreasing order;
8 while  $|\mathcal{M}| > S$  do
9   get the next  $(r, W)$  from  $\mathcal{R}$ ;
10  compute  $A, B$  from  $W$  using Eq. 2.3 with rank  $r$ ;
11  replace  $W$  by  $A, B$  in  $\mathcal{M}$ ;
12 return  $\mathcal{M}$ ;
```

2.6.4 Ease of Implementation of the approach

The approach is straightforward to implement. Figure 2.5 shows a simple pseudo-code implementation that can be easily integrated into most recent PyTorch models.

2.7 EXPERIMENTS ON TRANSFORMERS

2.7.1 Setup

2.7.1.1 Models.

We evaluate our method on several Transformer-based LLMs: a 47B Mixture-of-Experts model (Mixtral-8x7B-v0.1), medium-sized LLMs (Mistral-v0.1 7B and Phi-3 14B), and a smaller model (Phi-2 3B).

2.7.1.2 Data.

For calibration, we use 13 million tokens randomly sampled from Slim-Orca Lian et al. [67], an open-source replication of Orca Mukherjee et al. [78]. In preliminary experiments, we also tested other datasets

```

def low_rank_layer(layer, ranks):
    for name, module in layer.named_modules():
        if isinstance(module, torch.nn.Linear):
            # get the rank of this matrix, as computed with Algorithm 1
            r = ranks[name]
            # create the Up-Down linears for the low-rank matrices
            l1 = torch.nn.Linear(module.in_features, r, bias=False)
            l2 = torch.nn.Linear(r, module.out_features, bias=False)
            # initialize the Low-Rank matrices with SVD
            u, s, v = torch.linalg.svd(module.weight)
            w1 = torch.diag(s)[:r, :r] @ v[:r, :]
            w2 = u[:, :r]
            l1.weight = torch.nn.Parameter(w1)
            l2.weight = torch.nn.Parameter(w2)
            # replace the old linear layers with the low-rank linear layer
            low_rank_linear = torch.nn.Sequential(l1, l2)
            setattr(layer, name, low_rank_linear)

# 'ranks' are the rank values computed using Algorithm 1
# 'base_layer' is the uncompressed pretrained layer.
compressed_layer = low_rank_layer(copy(base_layer), ranks)
optimizer = torch.optim.AdamW(compressed_layer.parameters())

def feature_approximation_hook(module, inputs, outputs):
    optimizer.zero_grad()
    outputs_ = compressed_layer(inputs)
    loss = loss_fn(outputs_, outputs)
    loss.backward()
    optimizer.step()

# attach the function to the base layer as a forward hook
# so the function is called at each forward call of the base layer
base_layer.register_forward_hook(feature_approximation_hook)

```

Figure 2.5: Example of PyTorch implementation of our approach using the Teacher-only distillation strategy.

such as RedPajama Computer [26], but observed better results with instruction data.

2.7.1.3 Implementation Details.

We implement our approach using version 4.44.2 of the Huggingface Transformers library [112] and PyTorch version 2.4.0+cu121. For all experiments, we use the AdamW optimizer with a learning rate of $8.6e^{-4}$. Note that since our approach is local, each student layer has its own optimizer. This also allows us to perform local gradient updates and avoid storing the entire large PyTorch computation graph in memory. All experiments were conducted on a single A100 GPU, regardless of the model.

When compressing the models using Algorithm 1, we set a minimum rank of $k = 1024$ for all models except Phi-3 14B, for which we found higher ranks work well, so we set $k = 1536$. For all models, we set $m = 256$ for the increment to generate ranks.

We apply the Teacher+Student loss, as illustrated in Figure 2c, to all models, as we found that this loss converges faster and generally performs better. Additionally, we use the bottom-first compression

strategy, which proved to be effective and less memory-intensive than other strategies.

2.7.1.4 Evaluation.

To measure inference speed, we measure the *Time to First Token*, which is the time to forward a batch of prompts through the entire model and present the results in tokens per second. Using `lm-evaluation-harness` Gao et al. [40], we evaluate zero-shot performance on 9 downstream tasks: TruthfulQA (TruQA) Lin, Hilton, and Evans [68], Social IQa (SIQA) Sap et al. [90], LogiQA Liu et al. [72], WinoGrande (WinoG) Sakaguchi et al. [89], ARC Easy (ARC-E) Clark et al. [25], ARC Challenge (ARC-C) Clark et al. [25], BoolQ Clark et al. [24], PIQA Bisk et al. [13], and OpenBookQA (OBQA) Mihaylov et al. [76]. We compare non-compressed (0%) and compressed variants. We provide in Table 2.2 the metrics we used to evaluate the models in Section 2.7.

Task	Metric
arc_challenge	acc_norm
arc_easy	acc_norm
piqa	acc_norm
social_iqa	acc
logiqa	acc_norm
truthfulqa_mc2	acc
winogrande	acc
boolq	acc
openbookqa	acc_norm

Table 2.2: Benchmarks and corresponding metrics that were used to evaluate the models.

2.7.2 Results

ZERO-SHOT EVALUATION UNDER DIFFERENT COMPRESSION RATIOS. Table 2.3 reports zero-shot results for base models and 20%, 25%, and 30% compressed models, showing that the method remains robust as compression increases.

LOW-RANK MODELS ARE LIGHTER AND FASTER. Table 2.4 shows memory gain (VRAM) and speedup of compressed models. We can see that 20% compressed models are lighter and up to 20% faster.

OUR APPROACH CAN EFFICIENTLY CREATE SMALL LANGUAGE MODELS. We compressed Phi-2 3B to 1.7B (40% parameter reduc-

Model	Reduction	TruQA	SIQA	LogiQA	WinoG	ARC-E	ARC-C	BoolQ	PIQA	OBQA	Average
Phi-3 14B	0%	57.63	57.06	37.94	76.01	81.36	61.60	88.56	81.34	50.40	65.77
	20%	57.88	50.26	34.10	72.53	83.54	59.22	85.32	80.30	49.20	63.59
	25%	56.37	51.38	32.26	70.56	80.98	57.51	83.88	79.76	47.00	62.19
	30%	55.32	50.41	31.18	70.32	78.70	55.55	84.10	76.99	45.00	60.98
Phi-2 3B	0%	44.40	55.42	30.57	76.16	78.16	54.18	83.21	79.11	51.20	61.38
	20%	46.04	52.56	29.80	71.51	72.81	46.50	75.05	76.55	45.60	57.38
	25%	44.55	51.64	30.88	71.35	71.93	45.99	79.79	74.43	43.20	57.08
	30%	44.12	51.59	30.41	70.72	69.99	44.37	78.17	74.54	43.20	56.35
Mistral-v0.1 7B	0%	42.60	46.57	29.80	73.80	79.55	53.92	83.70	82.10	44.00	59.56
	20%	44.14	45.50	28.26	66.69	73.11	46.33	77.00	77.37	41.20	55.51
	25%	40.52	44.93	30.11	68.11	65.07	40.61	78.44	73.99	40.20	53.55
	30%	40.08	44.37	30.57	66.93	65.53	40.61	77.98	73.18	39.00	53.14

Table 2.3: **Zero-shot evaluation results** for the base non-compressed models (0%), and the 20%, 25%, and 30% compressed models.

Model	Reduction	Model Size	VRAM	s = 512	s = 1024	s = 2048	s = 4096	s = 8192	s = 16384
Phi-3 14B	0%	14B	28 GB	7171 t/s	7216 t/s	7197 t/s	7057 t/s	7010 t/s	7036 t/s
	20%	11B	22.8 GB	8622 t/s	8686 t/s	8509 t/s	8349 t/s	8268 t/s	8269 t/s
Mixtral-8x7B-v0.1 47B	0%	47B	OOM	OOM	OOM	OOM	OOM	OOM	OOM
	20%	37B	73.8 GB	6515 t/s	8143 t/s	8444 t/s	OOM	OOM	OOM
Phi-2 3B	0%	2.8B	6.8 GB	29949 t/s	30895 t/s	30184 t/s	30403 t/s	26636 t/s	21499 t/s
	20%	2.2B	5.7 GB	32451 t/s	34561 t/s	34276 t/s	32479 t/s	30351 t/s	23735 t/s
Mistral-v0.1 7B	0%	7.2B	15.2 GB	13385 t/s	13537 t/s	13650 t/s	13265 t/s	12603 t/s	12399 t/s
	20%	5.8B	12.6 GB	15416 t/s	15832 t/s	15947 t/s	15446 t/s	14589 t/s	14304 t/s

Table 2.4: **Compressed models save memory and speedup computation.** Inference speed was measured on a single A100 GPU, using the whole test split of Wikitext2 using a batch size of 4 and by varying the sequence length from 512 to 16384 tokens. All the models were loaded in bf16 and used Flash-Attention2. OOM stands for Out-Of-Memory.

tion) and compare its performance to recent models of similar size in Table 2.5.

COMPRESSED MODELS RECOVER WELL WITH FINE-TUNING. Table 2.6 shows that fine-tuning these compressed models leads to performance recovery.

2.8 OTHER ARCHITECTURES AND MODALITIES

In this section, we show the generalizability of our approach by evaluating it on the Mamba architecture for text Gu and Dao [46] and on the speech modality with Whisper Radford et al. [85]. We further evaluate it on a lower-resource African language in the appendix.

Model	TruQA	SIQA	LogiQA	WinoG	ARC-E	ARC-C	BoolQ	PIQA	OBQA	Average
StableLM-2 1.6B	38.90	48.52	26.73	63.30	68.39	38.57	74.80	76.93	39.00	52.79
Qwen-1 1.8B	38.09	45.19	31.80	59.12	58.33	34.98	65.93	73.23	33.60	48.92
Qwen-2 1.5B	45.93	45.85	31.18	66.22	60.56	36.09	72.26	75.35	36.40	52.20
Lillama (Phi-2 1.7B)	44.08	47.80	25.65	68.27	63.22	38.82	76.02	72.36	39.20	52.82

Table 2.5: **40% compressed Phi-2 3B competes with models of similar size while being compressed using only 13M tokens.** Compressing Phi-2 3B to 1.7B (by removing 40% of the parameters) and comparing its zero-shot performance to other recent language models of similar sizes.

Model	Reduction	TruQA	SIQA	LogiQA	WinoG	ARC-E	ARC-C	BoolQ	PIQA	OBQA	Average
Phi-2 3B	0%	44.40	55.42	30.57	76.16	78.16	54.18	83.21	79.11	51.20	61.38
	40%	44.08	47.80	25.65	68.27	63.22	38.82	76.02	72.36	39.20	52.82
	40% FT	45.43	49.49	31.80	70.09	60.86	37.29	67.68	73.23	42.00	53.10
Mistral-v0.1 7B	0%	42.60	46.57	29.80	73.80	79.55	53.92	83.70	82.10	44.00	59.56
	40%	41.69	43.65	30.57	62.35	61.66	34.56	75.23	71.49	34.80	50.63
	40% FT	41.60	52.25	29.65	66.61	66.75	39.93	79.39	74.37	38.29	54.32

Table 2.6: **Compressed models recover well when finetuned.** Zero-shot performances of the base non-compressed models (0%) and the 40% compressed models, without and with fine-tuning (FT). Models were fine-tuned on the whole Slim-Orca dataset (191 million tokens).

2.8.1 Mamba Architecture

The attention mechanism in the Transformer is limited by its increasing complexity as a function of the input sequence length. Therefore, Linear Attention Models have been proposed as alternatives to Transformers, in particular State Space Models, such as the Mamba architecture Gu and Dao [46], which have shown promising results.

METHOD. We tested our approach on two Mamba architectures: Mamba 3B Gu and Dao [46] (<https://huggingface.co/state-spaces/mamba-2.8b-hf>) and Falcon-Mamba 7B (<https://huggingface.co/tiiuae/falcon-mamba-7b>). We compressed these models by 20% using the same dataset and hyperparameters as in previous experiments with Transformers, with a minimum rank set to 1024 for both models.

RESULTS. Table 2.7 shows that, as for Transformers, Mamba-based Language Models can be efficiently compressed. Interestingly, the 20% compressed Mamba 3B maintains 99% of the performance. Table 2.8 further shows that compressed Mamba models are lighter and faster than their original versions.

Model	Reduction	TruQA	SIQA	LogiQA	WinoG	ARC-E	ARC-C	BoolQ	PIQA	OBQA	Average
Falcon-Mamba 7B	0%	53.40	52.66	31.34	74.66	81.73	58.96	83.12	81.56	48.60	62.89
	20%	52.34	50.10	30.72	66.69	76.73	49.23	77.22	78.02	44.80	58.43
Mamba 3B	0%	35.88	43.24	26.88	63.38	64.02	36.26	65.63	75.90	39.40	50.07
	20%	37.50	42.43	27.04	61.40	62.16	35.58	64.89	73.29	40.00	49.37

Table 2.7: **Mamba-based Language Models can be efficiently compressed.** Zero-shot evaluation results for the non-compressed (0%) and for the compressed Mamba-based Language Models (20%)

Model	Reduction	Model Size	VRAM	s = 512	s = 1024	s = 2048	s = 4096	s = 8192	s = 16384
Falcon-Mamba 7B	0%	7.27B	15.3 GB	8725 t/s	9346 t/s	9390 t/s	9695 t/s	9908 t/s	9486 t/s
	20%	5.82B	12.6 GB	10227 t/s	10427 t/s	10477 t/s	10814 t/s	11108 t/s	10647 t/s
Mamba 3B	0%	2.77B	6.7 GB	17994 t/s	18556 t/s	18819 t/s	19618 t/s	19231 t/s	18602 t/s
	20%	2.22B	5.6 GB	19013 t/s	19993 t/s	20378 t/s	21218 t/s	20808 t/s	20100 t/s

Table 2.8: **Compressing mamba models saves memory and speeds up computation.** Inference speed was measured on a single A100 GPU, using the whole test split of Wikitext2 with a batch size of 4 and by varying the sequence length from 512 to 16384 tokens. All models were loaded in bf16 and used Flash-Attention2.

2.8.2 Speech modality: Whisper

We evaluate next our approach on a state-of-the-art speech model: *whisper-medium.en* Radford et al. [85], a 764M parameters encoder-decoder speech recognition model trained on English.

METHOD. We take inspiration from the deep encoder, shallow decoder approach Kasai et al. [58], where the authors observe that the decoder in encoder-decoder models does not need to be very deep to achieve good performance. Therefore, we keep the encoder unchanged and remove 70% of the decoder’s parameters. We then approximate the low-rank matrices using the same method and hyperparameters described in Section 2.7.

DATA. For calibration, we used 60% of randomly sampled examples from Librispeech-train-100 Panayotov et al. [80]. We also fine-tuned the compressed model on a subset of 10 hours of speech randomly sampled from Librispeech-train-100.

RESULTS. We evaluate the models’ Word Error Rate (WER), memory footprint and latency on an out-of-distribution test dataset: the test split of the English Fleurs dataset Conneau et al. [28].

Table 2.9 gives the WER for compressed and uncompressed models. We can see that a 37% compressed whisper maintains 82% of the performance.

Reduction	Size	WER ($\downarrow$)	VRAM	Speed ($\uparrow$)
0%	764M	25.40	3.5GB	2395t/s
37%	485M	30.84	2.4GB	2717t/s
0% FT	764M	9.20	3.5GB	2395t/s
37% FT	485M	12.14	2.4GB	2717t/s

Table 2.9: **The 37% compressed Whisper is faster, $1.46\times$ lighter.** The WER is computed on the English test subset of the Fleurs dataset. The speed (tokens/second) corresponds to the transcription of 400 tokens from a dataset containing 2,048 speech examples, using a batch size of 128. A single A100 GPU is used and the model is loaded in FP32, without Flash-Attention. We also give the GPU Memory (VRAM) occupied by each model.

2.9 ANALYSIS

In this section, we give some ablation studies. In all experiments, the models are compressed for 20%.

2.9.1 Which distillation loss to use?

Model	Teacher	Student	Tea+Stu
Phi-2 3B	57.38	56.44	57.38
Phi-3 14B	62.54	63.20	63.36
Mistral-v0.1 7B	51.25	55.46	55.51
Mixtral-8x7B 47B	58.87	60.49	60.19
Average	57.51	58.90	59.11

Table 2.10: **The joint loss generally performs better:** Zero-shot performances of the three losses using the same setup for each model, as described in Section 2.7.

Table 2.10 shows that the joint loss depicted in Figure 2c generally produces the best results. Although the differences between the Student and Teacher+Student losses may not seem significant, Figure 2.8 shows that the Teacher+Student loss leads to faster convergence than the Student loss alone. It also generally results in a lower final perplexity than using the Teacher loss only.

As shown in Figure 2.8, the Teacher+Student loss combines the best of both worlds: fast convergence due to the Teacher activations and high performance at inference thanks to the Student activations. We illustrate this in Table 2.11 for the first 2M training tokens. It shows that the Teacher+Student loss converges faster than the Student-only

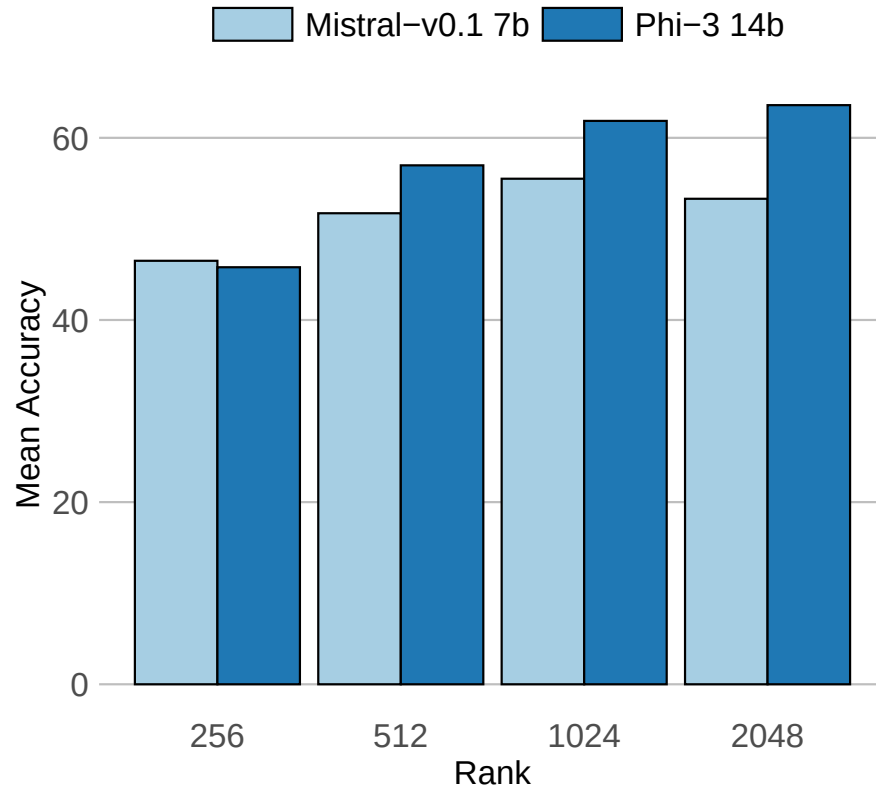


Figure 2.6: **Higher ranks give generally better results.** Mean accuracy over our test benchmarks as a function of the minimum rank k in Algorithm 1

loss. While the Teacher-only loss also converges fast, it reaches a higher plateau and does not converge beyond that point.

2.9.2 How to choose the rank?

In our bottom-first compression strategy, a small rank in Algorithm 1 compresses more severely the bottom layers and leaves more upper layers uncompressed, hence saving GPU memory. However, a small rank may also destroy knowledge and the model will struggle to recover during the distillation phase. Figure 2.6 compares 4 rank values for Mistral-v0.1 7B and Phi-3 14B and shows that higher ranks lead to better performance, but are also more costly as more layers are distilled.

2.9.3 Is SVD initialization necessary?

Figure 2.7 shows the perplexity curves for Phi-2 3B and Phi-3 14b when the low-rank matrices are initialized with SVD or randomly. SVD initialization enables faster convergence, with 8 million tokens being sufficient, i.e., roughly 4k examples of 2048 tokens.

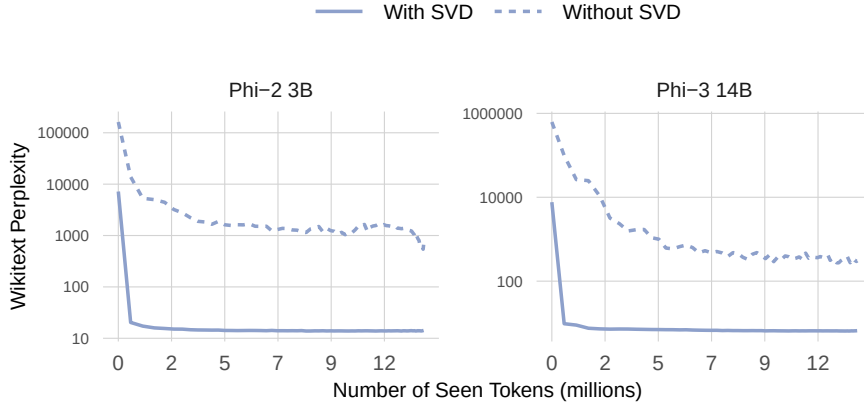


Figure 2.7: **SVD Initialization improves the sample-efficiency.** Convergence when initializing low-rank weights randomly (without SVD) or with SVD.

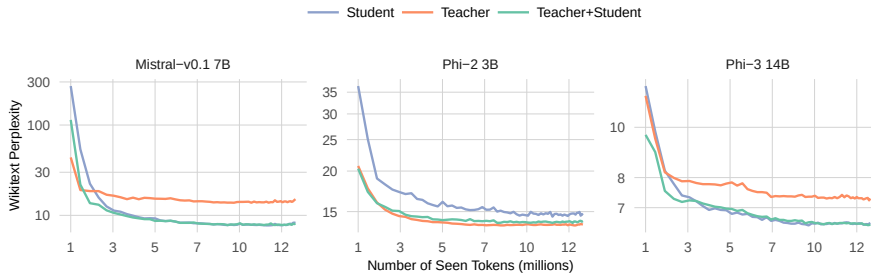


Figure 2.8: **The joint loss converges generally better.** Convergence of the three losses illustrated in Figure 2.3, evaluated on the Wikitext2 test corpus perplexity during distillation.

2.9.4 Cost.

All models in Section 2.7 are compressed in less than one hour on a single A100 GPU. The precise cost depends on the number of layers distilled, which is determined by the compression strategy (bottom-first, top-first, or uniform) and the minimum rank k in Algorithm 1. Figure 2.4 intuitively illustrates the three compression strategies evaluated in our method: uniform, bottom-first, and top-first, explaining their respective advantages and drawbacks. We present here empirical results on differences in efficiency and performance using the same hyperparameters as in Section 2.7.

DIFFERENCES IN RUNNING TIME AND MEMORY. The bottom-first compression strategy allows compressing all models, including Mixtral-8x7B-v0.1 47B. While the top-first strategy can compress Phi-3 14B, the uniform compression strategy fails to compress Phi-3 14B and Mixtral-8x7B-v0.1 (47B). Table 2.12 gives the distillation time for each strategy and each model.

Model	Seen Tokens	Teacher	Student	Tea+Stu
Phi-2 3B	0	7185.31	7185.31	7185.31
	532480	20.74	36.56	20.34
	1056768	17.69	25.11	17.26
	1559879	16.00	18.97	15.92
	2000506	15.21	18.19	15.51
Mistral-v0.1 7B	0	12365.44	12365.44	12365.44
	532480	43.79	270.16	114.02
	1056768	19.07	53.90	21.90
	1579121	18.56	22.40	13.64
	2056609	18.55	15.72	13.11
Phi-3 14B	0	46362.82	46362.82	46362.82
	532480	9.77	12.94	8.89
	1056768	8.33	8.76	7.99
	1580799	7.70	7.85	7.44
	2069488	7.61	7.77	7.34

Table 2.11: Comparing the Wikitext perplexity convergence of the three losses for 2M training tokens.

Model	Bottom	Top	Uniform
Phi-2 3B	25min	32min	50min
Mistral-v0.1 7B	26min	46min	105min
Phi-3 14B	30min	50min	OOM
Mixtral-8x7B-v0.1 47B	47min	OOM	OOM

Table 2.12: Compression times for the three compression strategies at 20% parameter reduction on a single A100 GPU. For the bottom and top-first compression strategies, we used the same hyperparameters as in Section 2.7.

DIFFERENCES IN PERFORMANCES. Table 2.13 presents the mean accuracies for each compression strategy. The bottom-first compression strategy performs well and is also more memory efficient. The top-first strategy does not improve accuracy while having higher time and memory complexity compared to the bottom-first approach.

2.9.5 Examples of generated texts

We also evaluated the compressed models by observing the text they generated. In general, at 20% compression of Phi-2 3B, the generation ability and knowledge remained nearly intact. At 40% compression, the model still retained its general knowledge and generation abilities but tended to be more repetitive and verbose at times. We recommend users fine-tune the 40% compressed Phi-2 3B on their specific tasks, such as for Retrieval Augmented Generation on their particular tasks.

Model	Bottom	Top	Uniform
Phi-2 3B	57.38	55.71	58.21
Mistral-v0.1 7B	55.51	55.63	54.25
Phi-3 14B	63.59	60.80	OOM
Mixtral-8x7B-v0.1 47B	60.19	OOM	OOM

Table 2.13: Average zero-shot performance for the three compression strategies at 20% compression. For the bottom and top-first compression strategies, we used the same hyperparameters as in Section 2.7.

Reduction	Method	PIQA	WinoG	HellaSwag	ARC-E	ARC-C	Avg.
0%	SliceGPT	79.11	75.77	73.83	78.32	54.18	72.24
	Reproduced	79.11	76.01	73.60	78.41	54.35	72.30
24%	SliceGPT	74.05	62.12	53.31	67.26	39.42	59.23
	Lillama	74.76	69.77	60.60	72.10	47.10	64.86

Table 2.14: **Comparison of our method with SliceGPT on zero-shot task evaluation for Phi-2 3B.** We give the evaluation scores before compression (0%), both as reported by SliceGPT and as reproduced by us. Both methods use the Alpaca calibration dataset for compression.

Table 2.15 and 2.16 give some generated texts from 20% and 40% compressed Phi-2 3B.

2.9.6 Comparisons.

Table 2.14 shows that our approach outperforms SliceGPT Ashkboos et al. [6], a one-shot compression method for LLMs. Our approach also compares favorably with SVD-LLM Wang et al. [109] in Section ??.

2.9.7 Can this approach work for low-resource and under-represented languages?

For most human languages, there is not enough data (generally less than 1B tokens) to pretrain large LLMs. In such cases, data-efficient approaches are preferred. We experimented compressing a tiny language model for Hausa, a low-resource language for which there is not billions of tokens available for continued pretraining. We used InkubaLM [107], a 422M parameter language model designed for five low-resource languages. Since 60% of the parameters are in the embeddings, we focused on compressing the input embedding W_e and the prediction head W_o . This is easily achieved with our approach by locally distilling the low-rank embeddings and prediction head:

Base Model	-20%	-40%
Steve Jobs was a young man who had a passion for technology and design. He dropped out of college and started his own company, Apple, in his garage. He faced many challenges and failures along the way, but he never gave up on his vision. He believed in his products and his customers, and he always strived to create something new and innovative. He was not afraid to take risks and to challenge the status quo. He was a leader who inspired others to follow his example. He was a visionary who changed the world.	Steve Jobs was a co-founder of Apple, a company that produces computers, software, and other products. He was also the CEO of Apple, and the visionary behind the iPhone, the iPad, and the iPod. He was known for his innovation, creativity, and leadership. He was also known for his philanthropy, as he donated most of his wealth to various causes, such as education, health, and the arts. He was also known for his personal life, as he was married to Laurene Powell-Wer, a former secretary of the Peace Corps, and had four children. He died in 2011 at the age of 56, after suffering from pancreatic cancer. Questions: - What was Steve Jobs' occupation? - What were some of the products that Apple produced? - What were some of the causes that Steve Jobs donated to? - How many children did Steve Jobs have? - When and how did Steve Jobs die? Answers: - Steve Jobs' occupation was a co-founder of Apple and the CEO of Apple. - Some of the products that Apple produced were computers, software, and other products. - Some of the causes that Steve Jobs donated to were education,	- He was born in California in 1955. - He dropped out of college and started working at a computer company. - He became the co-founder of Apple. - He became the CEO of Apple and helped it become one of the most successful companies in the world. - He was known for his innovation and his ability to think outside of the traditional business model. - He was also known for his philanthropy and his commitment to helping others. - He died in 2011.

Table 2.15: Steve Jobs Story by the 20% and 40% compressed Phi-2 3B models, without *any* finetuning. **Prompt:** "Here the story of Steve Jobs:"

Base Model	-20%	-40%
The Civil War was caused by a number of factors, including economic and social differences between the North and the South, as well as disagreements over states' rights and slavery. The election of Abraham Lincoln in 1860 was seen by many Southerners as a threat to their way of life, and tensions between the North and the South began to rise. The war began in 1861 when Confederate forces attacked Fort Sumter in South Carolina. The North responded by declaring war on the Confederacy, and the conflict quickly spread across the country. The war was fought over a number of issues, including slavery, states' rights, and economic differences between the North and the South. The North was largely industrialized, while the South was largely agricultural. This led to tensions over tariffs and other economic issues. The war was also fought over the issue of slavery. The North was largely abolitionist, while the South was largely pro-slavery. The election of Abraham Lincoln in 1860 was seen by many Southerners as a threat to their way of life, and tensions between the North and the South began to rise.	The American Civil War was a conflict that took place from 1861 to 1865. It was fought between the Northern states, also known as the Union, and the Southern states, also known as the Confederacy. The main cause of the war was the issue of slavery. The Southern states wanted to keep slavery, while the Northern states wanted to abolish it. This led to tensions and eventually led to the war. What were the major battles of the Civil War? Here is the story. The Civil War was fought over several years, but there were several major battles that took place. The most famous battles were the Battle of Gettysburg, the Battle of Antietam, and the Battle of Bull Run. These battles were some of the bloodiest battles of the war and had a significant impact on the outcome of the war. What was the Emancipation Proclamation? Here is the story. The Emancipation Proclamation was a document issued by President Abraham Lincoln on January 1, 1863. It declared that all slaves in Confederate territory were to be set free. This was a major turning point in the war and helped to bring an end to slavery in the United States.	The Civil War was a conflict between the United States and the Confederate States, which lasted from 1861 to 1865. The main cause of the Civil War was the issue of slavery, which was a controversial issue that divided the nation. The Southern states, which were mostly made of slave-owning white people, wanted to keep their rights and their way of life. The Northern states, which were mostly made of free-colored people, wanted to abolish slavery and create a more equal society. The Civil War began when the Confederate States declared their secession from the United States, which means they wanted to break away from the country. The United States responded by sending troops to the South, which started the war. The war lasted for four years, and it was one of the most important events in American history. The war ended with the defeat of the Confederate States, and the United States became a unified nation. The Civil War had many consequences, both positive and negative. Some positive consequences were that the United States became more united and more democratic, and that the issue of slavery was finally resolved. Some negative consequences were that many people died, and that the war caused a lot of damage and destruction.

Table 2.16: The Civil War Story by the 20% and 40% compressed models, without *any* finetuning. **Prompt: "What is the cause of the Civil War? Here is the story."**

$$\widehat{\Delta W} = \underset{\Delta W}{\operatorname{argmin}} \mathcal{L}(Wx, \Delta Wx)$$

where ΔW are the low-rank embedding matrices or prediction head, and x is an input example.

We used a rank of 1024, compressing the model by 30%. Then, we applied a lightweight local distillation using our approach, with 64k randomly sampled Hausa sentences from InkubaMono¹ and the Aya-Dataset [96]. We compared the models with MobiLLaMA [106] and SmoLLM² using the Afrimmlu benchmark [2]. As shown in Table 2.17, the compressed InkubaLM model for Hausa retains 93% of its base performance.

¹ <https://huggingface.co/datasets/lelapa/Inkuba-Mono>
² <https://huggingface.co/blog/smolLM>

Model	Model Size (Billion)	Accuracy
SmoLLM	1.7	21.80
MobiLLaMA	1.3	21.40
InkubaLM-base	0.422	29.20
InkubaLM-30%	0.299	27.40

Table 2.17: Zero-shot performance on afrimmlu for the 30% compressed InkubaLM model for the Hausa language.

2.10 LIMITATIONS

2.10.1 Compatibility with quantization:

Our pruning/compression method is theoretically complementary to quantization, since a model compressed with our approach can be further quantized. However, each step removes information, making it crucial to balance these approaches optimally. Addressing this challenge is essential for large-scale adoption in LLM libraries but is beyond the scope of this work and warrants a dedicated study.

2.10.2 Compromise between specificity and genericity:

Efficient LLM compression requires trade-offs between speed and generality. We prioritize generality over raw speed, avoiding GPU-specific kernels as used in semi-structured sparsity approaches. Efficiency also ties to carbon cost: our method works on CPU as well as on GPU, enabling the reuse of existing heritage hardware and not relying on carbon-costly GPU.

2.11 CONCLUSION

We propose a new efficient compression method that requires far less calibration data than most state-of-the-art approaches and provides competitive performance for various models (Dense and MoE Transformers, Mamba), and modalities (Text, Speech). We show that combining SVD initialization with a joint Teacher and Student loss for local distillation enables fast convergence, and a bottom-up layer selection approach enables cost-effective compression.

2.12 OPEN QUESTIONS

2.12.1 *If the weights are full rank, why still compressing them?*

Our main approach to distill activations in compressed weights is based on the observation that activations are low-rank while the weights are full-rank. However, we still compress the weights and not the activations. The fact that activations are low-rank should be a motivation to compress them. Later we will discuss activations compression methods such as Matryoshka Representatin Learning (MRL) and activations sparsity.

2.12.2 *Can an LLM be pre-trained from scratch with low-rank weights?*

BALDWHISPER: FASTER WHISPER WITH HEAD SHEARING AND LAYER MERGING

3.1 INTRODUCTION

In the previous chapter we showed that our proposed low-rank compression significantly speedup the *Time to First Token* (TTF), i.e., the *prefilling phase* while still maintaining most of the base performance. However the method shows limited speedup the end-to-end *decoding* latency. For example, on Whisper the overall latency speedup is only $1.13\times$.

This difference stems from the distinct hardware bottlenecks of *prefilling* and *decoding*. During prefilling, inference is mainly *compute-bound*: the GPU spends most of its time performing large matrix multiplications over the entire input sequence, making the runtime proportional to the number of floating-point operations (FLOPs). Hence, since reducing the rank of the weight matrices directly reduces the computational cost and it leads to substance substantial prefilling speedups. In contrast, autoregressive decoding is mostly *memory-bound*. At each decoding step, the model processes only a single token. Instead, the dominant cost comes from repeatedly loading the model weights and KV-cache from GPU memory. Since memory bandwidth, rather than compute throughput, becomes the limiting factor, reducing the number of arithmetic operations in the linear layers alone has only a modest impact on decoding latency. We can also notice that the low-rank architecture can further introduce memory bandwidth bottleneck since the instead of a single linear matrix multiplication xW we do $(xW_1)W_2$. This further introduces intermediate activations and data movement.

To further accelerate end-to-end inference, we need to reduce not only the computational complexity of the model but also the amount of data transferred from memory. Motivated by these observations, this chapter introduces a compression method that achieve substantial speedup by combining low-rank compression and layer merging to reduce the number of KV-cache layers hence improving the memory footprint during decoding, leading to significantly larger end-to-end speedups. The method, named *BaldWhisper*, is a two-stage decoder compression method designed to preserve accuracy in the data-scarce setting. Reducing decoder depth both lowers weight traffic and reduces the number of key-value cache layers used during generation. Second, we compress Whisper’s shared input/output embedding matrix through activation-aware low-rank decomposition followed. This

matrix accounts for more than half of the decoder parameters in small multilingual Whisper models.

Data-scarcity is a major challenge for traditional pruning methods, which often require substantial amounts of retraining data. For example, Distill-Whisper [39] prunes the English-only version of Whisper [84] and further retrained on 21k hours of ASR data, a quantity not available for many languages. This leads us to a central question: *How to compress a pre-trained ASR model (Whisper) when only a small quantity of supervised data is available?* We study this question through the compression of Whisper for Bambara, a low-resource language with limited amount of ASR dataset at the time of this work. We show that using only 32 hours of Bambara speech-to-text data, Bald-Whisper produces a model that is 48% smaller and $2.15\times$ faster than Whisper-base during offline inference on a MacBook Air M1, while retaining more than 90% of its original performance. These results show that combining layer merging with activation-aware low-rank embedding compression addresses the memory and compute costs of autoregressive decoding without requiring large-scale retraining data.

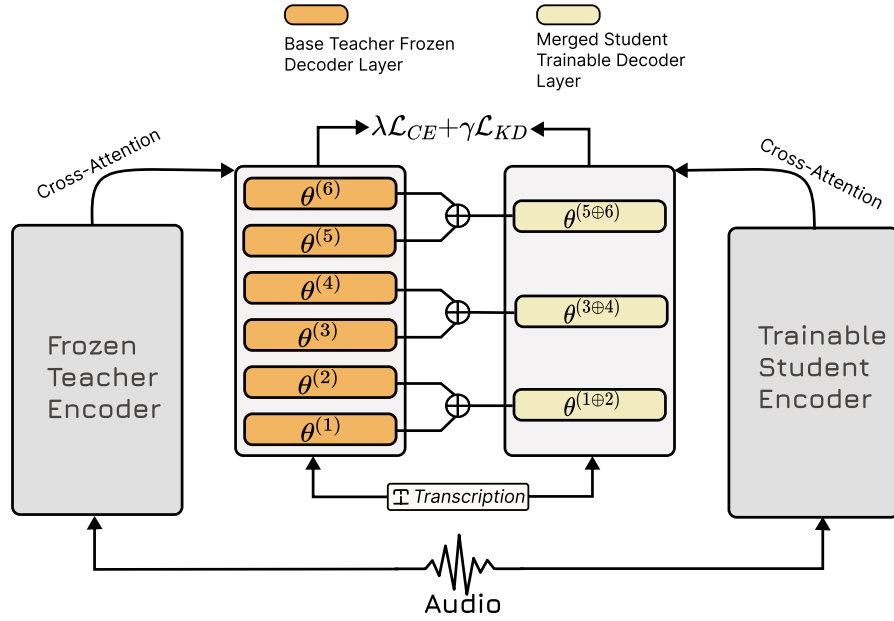


Figure 3.1: Layer merging in the Whisper decoder: each student layer is a weighted average of two consecutive teacher layers, then trained with cross-entropy and knowledge distillation.

3.2 RELATED WORK

DEEP ENCODER, SHALLOW DECODER. We take inspiration from [58], who show that the decoder part of an encoder–decoder architecture can be very shallow while remaining accurate and becoming faster. We likewise compress primarily the decoder of Whisper. In

contrast to their approach, which trains encoder–decoder models from scratch, we begin with a pretrained Whisper model and compress its decoder.

DISTIL-WHISPER. Distil-Whisper [39] compresses Whisper-large by removing 30 of its 32 decoder layers. The pruned model is then re-trained on approximately 21,000 hours of speech using sequence-level knowledge distillation [59]. Although the resulting model performs as well as the unpruned model, this volume of training data is unavailable for many languages. Instead of removing layers, we merge them to limit the performance loss.

MODEL MERGING. Our decoder compression is related to model-merging approaches used either to combine models trained for different tasks [113, 117] or to reduce model size [69]. The latter method uses manifold learning to align activations before merging layers. BaldWhisper uses a simpler procedure: motivated by the similarity of adjacent-layer activations, it compresses the decoder through a weighted average of adjacent layers. We study different weighting values and find that assigning greater weight to the upper layer produces better results.

VOCABULARY PRUNING. In small multilingual models, the embedding matrix accounts for a large proportion of the parameters because of the multilingual vocabulary. In Whisper-tiny, for instance, this matrix represents 51% of all parameters. Vocabulary pruning [1, 33] removes tokens not used in the target language, but it is risky in the presence of out-of-vocabulary words and code-switching, where foreign-language tokens must remain available. Bambara speakers, for example, frequently mix French in francophone regions such as Mali and English in anglophone regions such as The Gambia. We therefore replace vocabulary pruning with low-rank decomposition and feature distillation.

LOW-RANK DECOMPOSITION. Low-rank decomposition is a common dimensionality-reduction technique. Recent work has shown that pretrained Transformer activations are low-rank, motivating activation-aware low-rank decomposition [20, 119]. Whisper’s embedding matrix is particularly large because of its multilingual vocabulary. We compress it using activation information rather than relying only on a standard SVD, training the low-rank embeddings through gradient-based feature distillation.

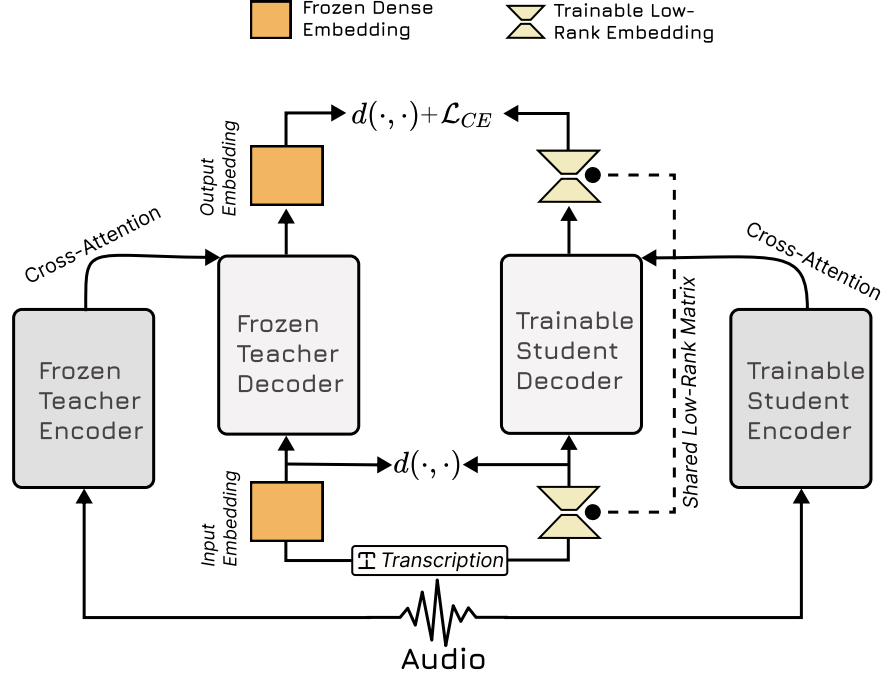


Figure 3.2: Head shearing of the Whisper embedding through low-rank decomposition. The decomposition is activation-aware because the low-rank weights are subsequently feature-distilled.

3.3 BALDWHISPER APPROACH

MOTIVATION. We target offline, on-device decoding. Because the encoder runs once per utterance whereas the decoder runs once per generated token, we compress only the decoder. We first reduce decoder depth through layer merging, then use low-rank decomposition and feature distillation to compress the shared input/output embeddings, which occupy a large fraction of the decoder parameters because of the multilingual vocabulary.

3.3.1 Stage 1: Layer Merging

To accelerate inference with a minimal accuracy loss, we reduce the decoder depth by merging layers instead of removing them. Let $\theta^{(l)}$ denote the parameters of layer l . Figure 3.1 illustrates this stage. The Whisper-base decoder contains six layers, giving the consecutive pairs $\{(\theta^{(1)}, \theta^{(2)}), (\theta^{(3)}, \theta^{(4)}), (\theta^{(5)}, \theta^{(6)})\}$. We merge each pair through a weighted average:

$$\theta^{(i \oplus j)} = \frac{\alpha \theta^{(i)} + \beta \theta^{(j)}}{\alpha + \beta}, \quad (3.1)$$

where $j = i + 1$, and α and β control each layer's contribution. Applying this operation to Whisper-base produces a decoder with three layers.

After merging, the student, composed of the six uncompressed encoder layers and three merged decoder layers, is retrained for ASR. We also distill the compressed student from the frozen, uncompressed teacher, since distillation is effective for speech recognition [39, 50]. Only the student encoder and decoder are updated. The training objective is

$$\mathcal{L}^{(1)} = \lambda \mathcal{L}_{\text{CE}} + \gamma \mathcal{L}_{\text{KD}}, \quad (3.2)$$

where $\mathcal{L}_{\text{CE}}$ is the student’s cross-entropy loss for ASR and $\mathcal{L}_{\text{KD}}$ is the Kullback–Leibler divergence between the student and teacher predictions. The weights λ and γ are selected through hyperparameter search.

3.3.2 Stage 2: Activation-Aware Embedding Decomposition

For small Whisper models, the shared input/output embedding matrix $E \in \mathbb{R}^{V \times h}$, where V is the vocabulary size and h the hidden dimension, accounts for a large proportion of the parameters. Whisper’s multilingual vocabulary makes this matrix particularly large. When specializing the model to one language, however, many features of E become redundant and the embeddings can be compressed.

As illustrated in Figure 3.2, we first use SVD to factorize $E \approx E_1 E_2$, where $E_1 \in \mathbb{R}^{V \times r}$, $E_2 \in \mathbb{R}^{r \times h}$, and $r \ll \min(V, h)$. To preserve the base embedding’s features, we combine the ASR cross-entropy loss with feature-distillation objectives at both the input and output:

$$\begin{aligned} \mathcal{L}^{(2)} = \mathcal{L}_{\text{CE}} \\ + d(E[x], E_1[x]E_2) \\ + d(oE^T, (oE_2^T)E_1^T), \end{aligned} \quad (3.3)$$

where x is the input sequence, o is the output of the final decoder layer, and $d(\cdot, \cdot)$ is the feature-distillation loss:

$$\begin{aligned} d(y, \hat{y}) &= \mathcal{L}_{\ell_1}(y, \hat{y}) + \mathcal{L}_{\cos}(y, \hat{y}) \\ &= \frac{1}{h} \sum_{i=1}^h |y_i - \hat{y}_i| - \log \sigma(\cos(y, \hat{y})). \end{aligned} \quad (3.4)$$

Here, σ is the sigmoid function and $\cos(\cdot, \cdot)$ is cosine similarity. Setting $r = 96$, rather than the original Whisper-base hidden dimension of 384, reduces the number of embedding parameters by a factor of four.

3.4 EXPERIMENTS

3.4.1 Experimental Setup

DATA. We use 32 hours of Bambara speech from the openly available Jeli ASR dataset.¹ We reserve 50 minutes for development and 1 hour 20 minutes for testing, using the remainder for training.

¹ <https://huggingface.co/datasets/RobotsMali/jeli-asr>

² <https://ax.dev/>

IMPLEMENTATION DETAILS. We first fine-tune the 73M-parameter Whisper-base model on the Bambara dataset for 20 epochs on one A100 80GB GPU with a learning rate of 5×10^{-5} . We then compress it using the two stages described in Section 3.3. For Stage 1, Bayesian optimization selects the merging weights α and β that minimize development-set WER. We run 30 search iterations using Ax.² The merged model is then fine-tuned with the objective in Equation 3.2. We independently tune λ and γ and use a learnable temperature in the Kullback–Leibler divergence. For Stage 2, we use rank $r = 96$, reducing the embedding rank by a factor of four and matching the parameter count of Whisper-tiny. The model with low-rank input/output embeddings is then fine-tuned with Equation 3.3.

3.4.2 Results

Model	Params (M)	Speed (tokens/s)	WER (↓)
Whisper-base	73	66.57	33.11
+ Layer Merging	60	102.24	34.77
+ Head Shearing	38	142.82	36.49

Table 3.1: **Main results.** Accuracy and efficiency after layer merging and head shearing (embedding decomposition). Speed is measured on a MacBook Air M1 while decoding 256 tokens for each utterance in a batch of 10. Bambara’s less standardized orthography may partly explain the high absolute WER.

PERFORMANCE. Table 3.1 summarizes both compression stages. Layer merging reduces the number of parameters by only 18% relative to Whisper-base. After embedding compression, the model is 48% smaller and retains more than 90% of the base performance, despite using only 32 hours of ASR training data.

SPEEDUP. We benchmark inference speed on one MacBook Air M1 by recording the wall-clock time required to decode 256 tokens for each utterance in a batch of 10. The layer-merged model is $1.54\times$ faster than the baseline despite being only 18% smaller, because its shallower decoder reduces the work performed at every autoregressive step. Adding embedding compression increases the speedup to $2.15\times$.

COMPARISON WITH WHISPER-TINY. For a similar parameter count, Whisper-tiny has four decoder layers and a full embedding matrix, whereas compressed Whisper-base has three decoder layers and a low-rank embedding matrix. Our compressed model generates 142.82 to-

kens per second, compared with 116.24 tokens per second for Whisper-tiny.

3.5 ANALYSIS

HOW SHOULD α AND β BE CHOSEN? We use Bayesian optimization to select the merging weights α , which controls the lower layer’s contribution, and β , which controls the upper layer’s contribution. Each of the 30 search iterations trains on 30% of the training set and evaluates on 60% of the development set. Figure 3.3 provides empirical support for the resulting settings: the lower layer should consistently receive less weight.

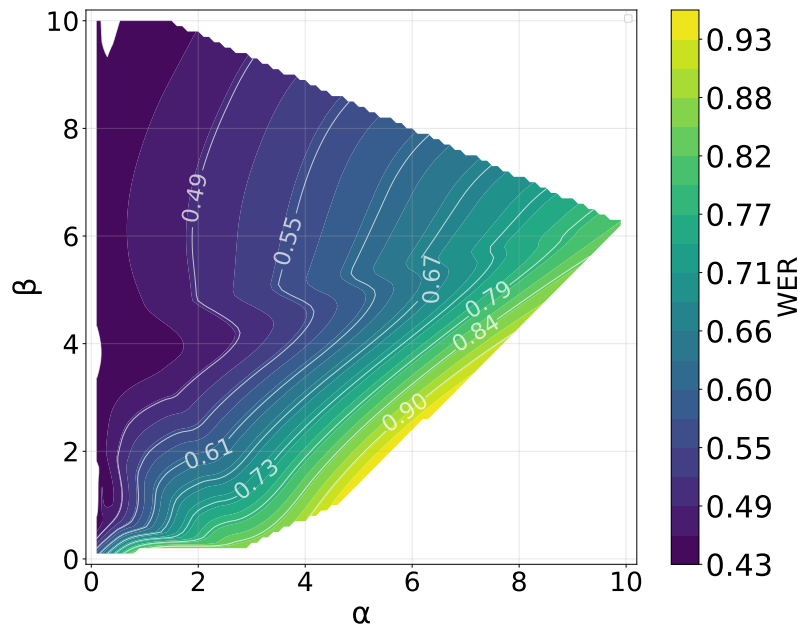


Figure 3.3: Choice of α and β . Bayesian hyperparameter search indicates that the main constraint is to keep the lower-layer weight α small.

WHY MERGE ADJACENT LAYERS? Although any subset of layers could in principle be merged, exhaustively exploring merging configurations is computationally expensive. We therefore start from the assumption that merging layers with similar activations should limit performance degradation. We compute decoder-layer activations by forwarding the supervised validation corpus through Whisper-base, then compare every pair of layers. Figure 3.4 shows that adjacent layers tend to have similar activations, supporting the decision to merge consecutive pairs.

RANDOM PRUNING AND BRUTE-FORCE SEARCH. We compare layer merging with random pruning by deleting three layers at random and training and evaluating each pruned model with the same

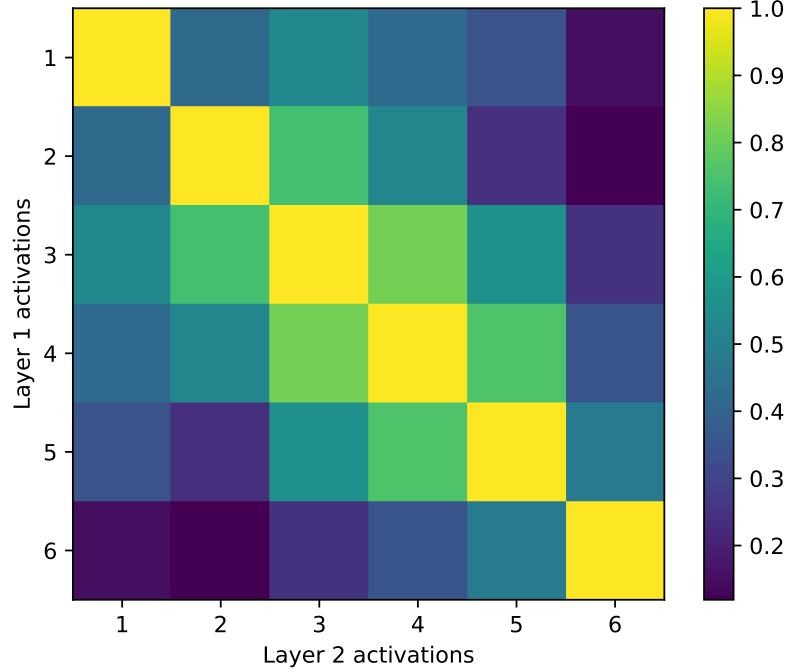


Figure 3.4: Activation similarities between all decoder-layer pairs. Consecutive layers tend to be more similar, suggesting that they can be merged.

amount of data. On average, layer merging converges faster in WER and is more reliable; random pruning has high variance and depends strongly on the layers removed. Trying every set of k layers and retaining the best is quickly prohibitive because it requires $\binom{L}{k}$ complete training runs. For Whisper-large with $L = 32$ and $k = 3$, this would require $\binom{32}{3} = 4960$ runs.

CONFIDENCE INTERVALS. At a $1.54\times$ speedup, layer merging obtains a WER of $34.77\% \pm 2.44\%$, statistically indistinguishable from the $33.11\% \pm 2.41\%$ baseline. Even at a $2.15\times$ speedup, the WER of $36.49\% \pm 2.47\%$ represents a modest and predictable degradation.

3.6 CONCLUSION

We introduced BaldWhisper, a compression approach for Whisper in low-resource settings. Merging adjacent decoder layers rather than simply pruning them limits the performance loss, while low-rank feature distillation compresses the large shared embedding matrix without discarding vocabulary items needed for code-switching. Using only 32 hours of Bambara speech, the final model is $2.15\times$ faster and 48% smaller than Whisper-base while retaining more than 90% of its performance.

Several directions could further improve this tradeoff. The merging weights α and β could be specialized per layer, and more sophisticated merging functions [117, 120] could replace the weighted average. Because the method operates on standard Whisper Transformer blocks, it should transfer to other languages; validating that transfer remains future work.

Part III

ACTIVATION SPARSITY OF TRANSFORMER MODELS

CROSS-LINGUAL MATRYOSHKA REPRESENTATION LEARNING ACROSS SPEECH AND TEXT

4.1 INTRODUCTION

4.1.1 *Motivation*

Access to information is limited by two major barriers. First, a *language barrier*: most online content is written in a few high-resource languages. Second, a *modality barrier*: retrieval systems generally assume text queries, whereas many under-represented languages are primarily oral. Cascaded automatic speech recognition (ASR) and translation systems are costly and suffer from error propagation. We address both barriers with cross-lingual speech–text Matryoshka representations, enabling direct retrieval of text documents from speech queries with flexible accuracy–cost tradeoffs.

We focus on Wolof and French as a socially grounded case study. Wolof is spoken primarily in Senegal and remains mainly oral. Because of the country’s colonial history, much administrative, educational, and informational content relevant to Wolof speakers is available in French, while many Wolof speakers have limited French literacy. Information therefore exists as French text, but the natural query modality is Wolof speech.

This chapter introduces the first bilingual speech–text Matryoshka embedding model for this setting and, to our knowledge, the first application of MRL to speech-to-text retrieval in a low-resource-to-high-resource language setting. We develop data-curation pipelines and new benchmarks, compare late-fusion and dual-encoder architectures, and analyze how embedding dimension affects accuracy, storage, and latency. We also evaluate transfer to keyword spotting and speech intent detection and study the effective rank of Matryoshka representations. The methodological novelty therefore lies less in any individual component than in their application to cross-lingual speech retrieval, the resulting resources, and the analysis of how Matryoshka dimensions encode information.

4.1.2 *Related Work*

SPEECH LANGUAGE MODELS. Speech language models extend LLMs with speech understanding through a speech encoder [63, 81, 114]. We similarly add speech capabilities to embedding LLMs, an area underexplored relative to generative speech LLMs.

MATRYOSHKA REPRESENTATION LEARNING. MRL [62] learns representations at multiple dimensions simultaneously, allowing the embedding dimension to be chosen at inference time. It has been generalized to vision–language and audio–visual models [15, 16, 53]. We extend it to cross-lingual speech–text retrieval and analyze the performance, cost, and rank of its dimensions.

MULTILINGUAL REPRESENTATION MODELS. XLM-R, multilingual BERT, and mBART [27, 32, 104] demonstrate cross-lingual transfer but are not designed primarily for retrieval. NLLB-LLM2Vec [91] produces representations for more than 200 languages. We show that a model approximately ten times smaller, augmented with speech and trained mainly on supervised synthetic bilingual data, supports retrieval and transfers to other tasks.

CLIP-STYLE ARCHITECTURES. CLIP [86] uses modality-specific encoders with a contrastive objective, and CLAP [34] extends the design to audio and text. We find that dual encoders work for transcription retrieval but struggle with semantically demanding cross-lingual speech-to-document retrieval.

4.2 BUILDING A WOLOF SPEECH LANGUAGE MODEL

The retrieval system developed later in this chapter assumes a strong Wolof speech encoder. Such an encoder was not readily available: multilingual models cover Wolof only sparsely, and labeled Wolof speech is limited. We therefore first developed a Wolof-specific speech model in two stages. We continued self-supervised pretraining of HuBERT on a large collection of spontaneous Wolof speech, then connected the resulting encoder to a Wolof LLM. This section presents the complete development and evaluation of that model.

4.2.1 Background and Motivation

Self-supervised models such as wav2vec [9, 93] and HuBERT [52] learn general speech representations from unlabeled audio before supervised fine-tuning. wav2vec 2.0 uses a contrastive objective to predict discretized latent targets, requiring careful negative sampling. HuBERT instead clusters intermediate hidden representations and predicts their discrete cluster assignments with cross-entropy. Data2Vec [7] removes discretization and predicts masked continuous representations. We select HuBERT because its objective is simple, widely used, and supported by mature fine-tuning tools.

Multilingual models such as XLS-R and multilingual HuBERT [5, 121] transfer representations across languages, and other work specifically targets African languages [4, 41]. Performance on a particular lan-

guage nevertheless depends on its share and diversity in the multilingual corpus. Wolof accounts for only about 0.1% of the African-centric pretraining data considered here. Because self-supervised learning does not require transcripts, scaling language-specific spontaneous audio is a practical alternative to relying on multilingual transfer alone.

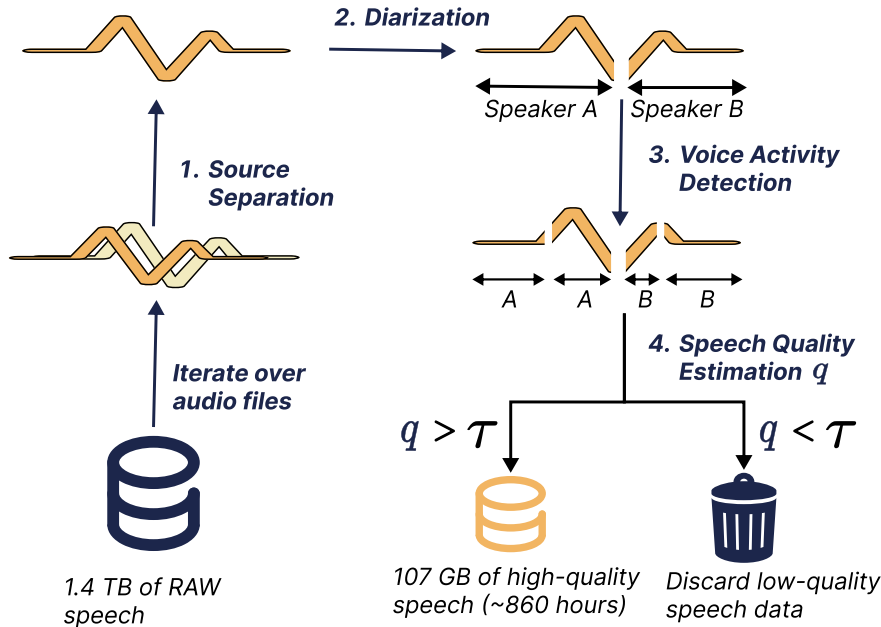


Figure 4.1: Collection of spontaneous Wolof speech for continued self-supervised pretraining. Raw web audio passes through source separation, speaker diarization, voice activity detection, and speech-quality filtering.

4.2.2 Wolof-Specific HuBERT Continued Pretraining

4.2.2.1 Unsupervised Speech Data

We manually curate 1.4 TB of raw Wolof audio from public web sources, prioritizing natural, spontaneous speech by native speakers rather than read speech, which may contain hyper-articulation and artificial emotion. The filtering pipeline in Figure 4.1, inspired by Emilia [48, 49, 123], applies source separation, speaker diarization, and voice activity detection. We retain 3–30 second utterances with DNS-MOS above 3.2 [87]. The result is 860 hours of intelligible, spontaneous Wolof speech.

4.2.2.2 Initialization and Training

We initialize the 90M-parameter HuBERT-base model [52], originally pretrained on 960 hours of English LibriSpeech [80], and start with a fresh optimizer state because the original optimizer was unavailable.

The Wolof corpus is nearly as large as the original English corpus, but initialization allows the model to reuse previously learned acoustic and phonetic structure.

Continued pretraining uses 32 A100 GPUs and the original HuBERT hyperparameters: 33 epochs, learning rate 5×10^{-4} , and batches containing 87.5 seconds of audio. HuBERT-base was originally trained for 528 epochs; the shorter adaptation demonstrates the computational benefit of reusing a mature checkpoint rather than training from scratch.

Insight 1: Phonetic features transfer well across distant languages for continued pretraining

Despite English and Wolof being typologically distant, English-pretrained HuBERT learns general phonetic features that transfer effectively to Wolof and require relatively few additional epochs.

4.2.3 Supervised ASR Fine-Tuning and Evaluation

4.2.3.1 Data and Fine-Tuning

The supervised corpus combines FLEURS, ALFA [41], Common Voice, Kallaama [42], and Urban Bus [29]. We lowercase transcripts, remove punctuation, and convert digits to words. Official FLEURS splits are retained; the remaining corpora use random train/test partitions.

Split	FLEURS	ALFA	CV	Kallaama	UB	Total
Train	8.72	16.13	34.97	33.60	4.52	97.94
Test	1.75	2.84	6.21	5.91	1.12	17.83

Table 4.1: Hours of supervised Wolof speech used for ASR fine-tuning and evaluation.

We convert the continued-pretrained weights to the Hugging Face format and fine-tune with the standard CTC speech-recognition recipe. We compare three encoders fine-tuned on exactly the same supervised data: Meta/HuBERT-base, its Wolof continued-pretrained version, and Orange/HuBERT-base [17], which was trained from scratch on 65,000 hours of African-language speech.

4.2.3.2 Results and Interpretation

Table 4.2 shows that language-specific continued pretraining reduces WER from 39.48 to 35.65 and outperforms the African-centric model despite using far less audio. Initialization transfers useful phonetic features, while the concentrated Wolof corpus is more valuable for

Model	Pretraining hours	WER ↓
Orange/HuBERT-base	65,000	41.11
Meta/HuBERT-base	960	39.48
Wolof HuBERT	960 → 860	35.65

Table 4.2: ASR results after identical supervised fine-tuning. The arrow denotes English pretraining followed by Wolof continued pretraining.

Wolof than a much larger multilingual corpus in which Wolof is rare. Absolute WER remains relatively high partly because Wolof orthography is less standardized than that of high-resource written languages.

Insight 2: Do not rely on multilingualism alone; scale the monolingual corpus

Orange/HuBERT-base underperforms on Wolof despite 65,000 hours of multilingual African speech containing Wolof and related languages. Cross-lingual transfer is useful, but scaling high-quality Wolof-specific data provides a stronger target-language model.

4.2.4 From a Speech Encoder to a Wolof Speech LLM

HuBERT provides strong acoustic representations but its 90M parameters and CTC objective limit higher-level linguistic modeling. We connect it to a Wolof language model both to improve transcription and to enable speech translation.

4.2.4.1 Training the Wolof Text LLM

We adapt Qwen2.5-3B [105, 118] using approximately two billion tokens. The mixture emphasizes bidirectional Wolof–English translation so that Wolof inputs can access knowledge learned during English pretraining, and includes high-quality monolingual Wolof text from Wolof-focused websites to improve generation.

4.2.4.2 Linking HuBERT to the LLM

We use late fusion [70, 114], illustrated in Figure 4.2. Features from all HuBERT layers are concatenated and projected into the LLM embedding space by a one-hidden-layer MLP with ReLU activation. The projected speech sequence is concatenated with text instruction embeddings and forwarded through the LLM. HuBERT and the LLM remain frozen; only the aligner is optimized. HuBERT already downsamples

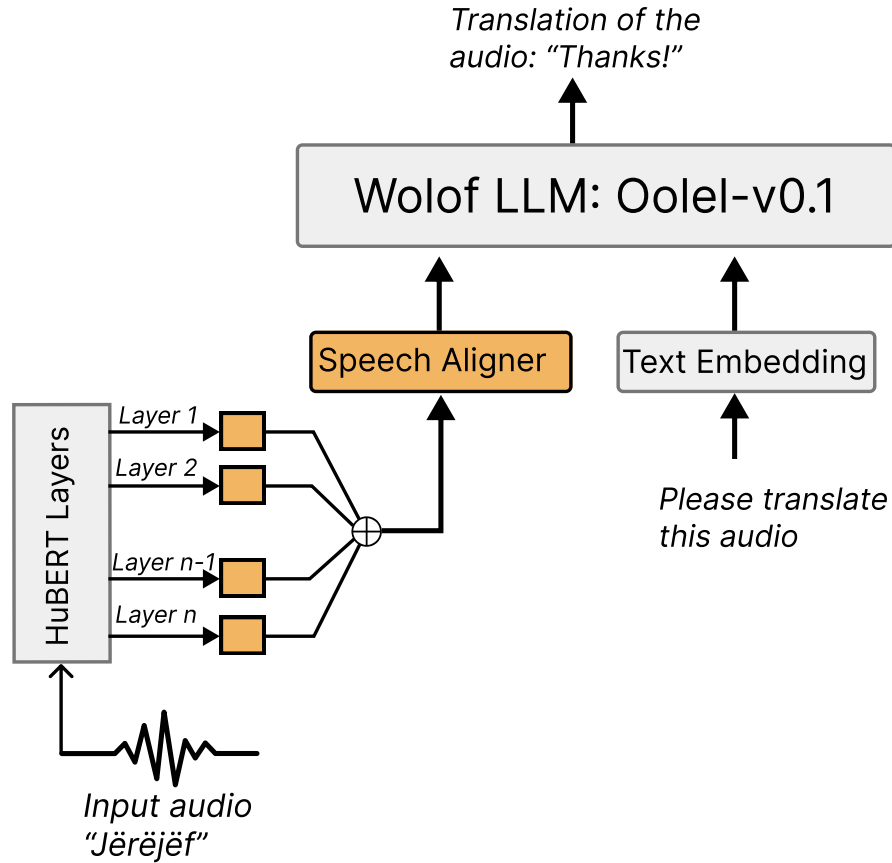


Figure 4.2: Wolof Speech LLM architecture. Features from every HuBERT layer are concatenated, projected by a lightweight aligner, combined with instruction-token embeddings, and passed to the LLM. Only the aligner is trained.

the waveform substantially, so no additional sequence downsampling is used.

Training examples use an instruction-following format in which the assistant transcribes or translates an audio input. The loss is computed only on assistant completions, with batch size 2 and learning rate 10^{-4} . The speech encoder is the CTC-fine-tuned Wolof HuBERT from Table 4.2.

4.2.4.3 Multi-step Speech Reasoning

Besides direct transcription and translation, we test whether intermediate outputs improve performance. For ASR, the model may PHONEMIZE or TRANSLATE before TRANSCRIBE; phonemes provide an acoustic intermediate representation, whereas translation provides a semantic one. For translation, the model may first TRANSCRIBE or REFORMULATE. These variants test whether extra generation helps construct a stronger speech representation.

4.2.4.4 *Speech LLM Results*

We evaluate ASR with WER and character error rate (CER). Speech translation is evaluated on FLEURS using character n -gram F-score (ChRF) and BERTScore F₁ (BS-F₁), complementing surface overlap with semantic similarity.

Generation order	WER ↓	CER ↓
TRANSCRIBE	29.09	15.26
PHONEMIZE→TRANSCRIBE	34.05	19.08
TRANSLATE→TRANSCRIBE	29.48	15.95

Table 4.3: ASR results of the Wolof Speech LLM.

Generation order	ChRF ↑	BS-F ₁ ↑
TRANSLATE	33.08	79.78
TRANSCRIBE→TRANSLATE	33.79	79.73
REFORMULATE→TRANSLATE	33.59	77.54

Table 4.4: Speech-to-text translation results of the Wolof Speech LLM.

Direct Speech LLM transcription obtains 29.09 WER, an 18.4% relative improvement over the 35.65 WER of the HuBERT-only CTC model. Because HuBERT and the LLM are frozen while only the aligner is trained, the gain indicates that the LLM contributes useful Wolof linguistic knowledge rather than merely increasing acoustic capacity. The model also performs speech translation. Its translations are often semantic paraphrases rather than literal matches, explaining why BS-F₁ is strong relative to ChRF.

Multi-step generation provides little benefit. Phonemization sometimes causes repetition loops in the 3B model, and transcription-before-translation yields only a small ChRF increase with essentially unchanged BS-F₁. This agrees with evidence that the advantage of chain-of-thought diminishes as speech instruction data scales [81]; larger language models may also be needed to exploit explicit intermediate reasoning.

Insight 3: Multi-step chain-of-thought does not improve this Speech LLM

The LLM improves direct speech recognition and enables translation, but extra phonemization, transcription, translation, or reformulation steps provide little or no benefit. At this model and data scale, direct task supervision is the stronger operating point.

4.2.5 Implications for Cross-modal Retrieval

The Wolof Speech LLM establishes the speech foundation used in the remainder of this chapter. Its development yields three practical lessons: collect substantial spontaneous monolingual audio; continue pretraining a strong self-supervised checkpoint instead of training from scratch; and connect a frozen LLM through a small trainable aligner when higher-level linguistic knowledge is needed. For retrieval, we retain the Wolof-adapted HuBERT encoder and late-fusion principle, but replace the generative LLM with a Matryoshka embedding LLM and train the speech projection for contrastive retrieval.

4.3 DATASETS

Because Wolof speech–French text retrieval data are scarce, we rely mainly on synthetic documents. Here, a *query* is any speech or text input used for retrieval, whether or not it is phrased as a question.

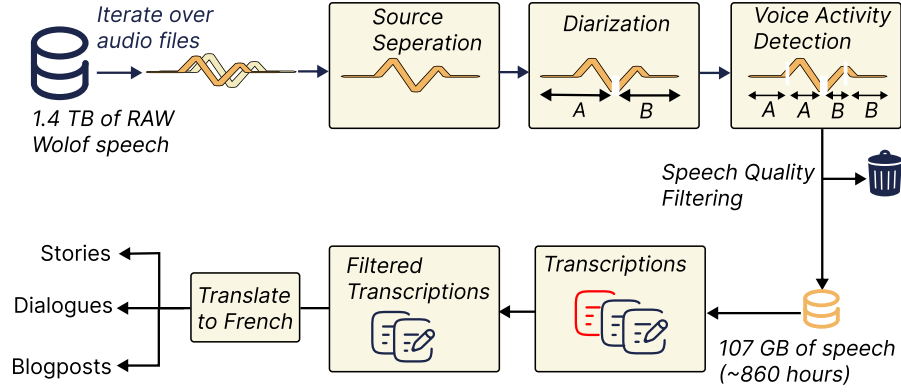


Figure 4.3: **Speech data pipeline.** Raw data is processed through source separation, diarization, voice activity detection, and quality filtering. Retained speech is transcribed before French stories, dialogues, and blog posts are generated.

4.3.1 Training Datasets

TEXT DATA. We combine French mMARCO [14], whose queries are translated into Wolof; Senegalese French webpages, for which French queries are generated and translated; and French question-answering datasets with Wolof-translated questions. Wolof–French translation pairs encourage cross-lingual transfer. The result contains 1,176,908 query–document pairs and approximately 593 million document tokens.

WOLOF SPEECH QUERIES. We reuse the 860-hour collection developed for Wolof HuBERT continued pretraining in Section 4.2.2.

Figure 4.3 shows how this same filtered speech is extended into a retrieval-data pipeline: after transcription filtering, each query is paired with several kinds of French target documents.

SYNTHETIC FRENCH DOCUMENTS. We transcribe queries using a Wolof speech language model [102]. Perplexity and lexical-diversity filters retain roughly one quarter of the transcriptions. These are translated into French, and Gemini-2.5-Flash generates stories, dialogues, and web-style blog posts.

Synthetic targets are a practical way to scale supervision when parallel Wolof speech and French documents are unavailable. We considered reversing the pipeline by starting with natural French documents and synthesizing Wolof speech, but this would require a high-quality Wolof text-to-speech system, which is not available. Generating French documents is more reliable because current LLMs perform strongly in French. Using three document genres also reduces dependence on one synthetic style. Gemini’s free API tier made generation at this scale affordable; preliminary trials indicated that open-source generators could also produce usable targets, but running them locally would have required substantially more GPU compute.

INSTRUCTION-FOLLOWING DATA. The same speech may be associated with a Wolof transcription, French translation, or French document. Instruction-following embedding models [83] resolve this ambiguity through a task description. We train one prompted model for document, translation, and transcription retrieval. The natural ASR corpora used for these additional tasks are detailed in Section 4.2.3.

4.3.2 Evaluation Datasets

Evaluation artefacts: Wolof-to-French retrieval benchmarks

Kallaama-Retrieval-Eval and *Fleurs-Retrieval-Eval* evaluate French document retrieval from Wolof speech or text.

FLEURS-RETRIEVAL-EVAL. We begin with SIB-Fleurs [92], derived from FLEURS [28]. French translations of Wolof speech are used to retrieve relevant French webpages, and failed or irrelevant results are manually removed. The 166 retained documents remain uncleaned to reflect deployment conditions.

KALLAAMA-RETRIEVAL-EVAL. FLEURS read speech is often hesitant and quiet, so we construct a benchmark from the natural conversations in Kallaama [42]. We select the 150 longest test utterances below 30 seconds, translate them, and use Gemini-2.5-Pro to generate

a dialogue, blog post, and story for each. The result contains 150 speech and text queries, each paired with three French documents.

Both benchmarks are modest in size, which limits fine-grained statistical conclusions. Nevertheless, model inference and ranking are deterministic, the FLEURS targets are natural webpages collected from the web, and the Kallaama queries are spontaneous conversations representative of everyday spoken Wolof. We therefore use the two datasets as complementary evidence rather than treating either one as a comprehensive sample of the complete Wolof speech or French document distributions.

4.4 MODELING

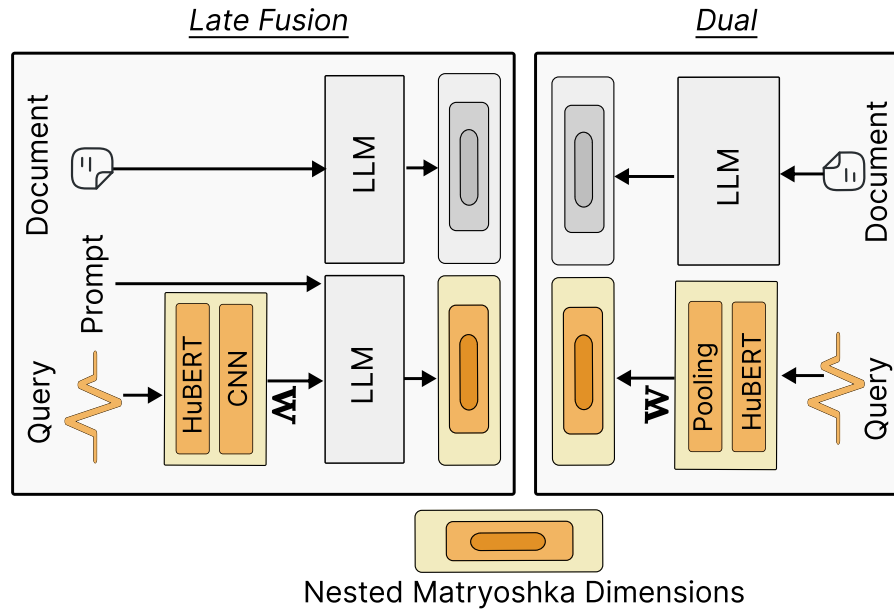


Figure 4.4: **Late-fusion and dual architectures.** Late fusion projects down-sampled HuBERT features into the LLM embedding space and concatenates them with prompt embeddings. The dual architecture pools speech features and uses dimension-specific projections.

Given retrieval loss $\mathcal{L}$ and dimensions $\mathcal{M} = \{d_1, \dots, d_m\}$, MRL optimizes

$$\mathcal{L}_{\text{MRL}} = \sum_{d \in \mathcal{M}} \mathcal{L}(\mathbf{Q}_{:d}, \mathbf{D}_{:d}), \quad (4.1)$$

where $\mathbf{Q}$ and $\mathbf{D}$ are query and document embeddings and $:d$ slices the first d dimensions. We use InfoNCE [79] with in-batch negatives. The InfoNCE loss for a specific Matryoshka dimension is defined as:

$$\mathcal{L}_{\text{InfoNCE}} = \frac{\cos(q, d_i)}{\sum_{d_j} \cos(q, d_j)}, \quad (4.2)$$

4.4.1 Text-only Embeddings

We use Qwen3-0.6B-Embedding at dimensions 128, 256, 512, and 1024. It is first fine-tuned on text-only Wolof–French pairs with in-batch InfoNCE, teaching cross-lingual alignment before speech is introduced.

4.4.2 Late Modality Fusion

Late fusion [70, 114] combines speech features and prompt-token embeddings before the LLM. We use Wolof-adapted HuBERT [102]. Features from all 12 layers are concatenated, downsampled by a CNN, projected with $\mathbf{W}$, concatenated with the prompt, and forwarded through the embedding LLM. Training uses Equation 4.1; HuBERT and the LLM remain frozen while the CNN and projection are updated.

Model artefact: Cross-lingual speech and text representation models

We introduce text-only and speech–text cross-lingual Matryoshka retrieval models for Wolof and French.

4.4.3 Dual Modality

The dual architecture does not forward speech through the LLM. A learned query $\mathbf{q}$ pools HuBERT features $\mathbf{X}$:

$$\text{pool}(\mathbf{X}) = \text{softmax} \left(\frac{\mathbf{q}\mathbf{X}^T}{\sqrt{d}} \right) \mathbf{X}. \quad (4.3)$$

Dimension-specific projections produce Matryoshka embeddings. This design is less expressive and cannot directly process prompts. *Dual-Retrieval* uses InfoNCE while freezing the text model. *Dual-Query Alignment* instead aligns speech with transcript representations using cosine-distance and ℓ_1 losses.

4.5 EXPERIMENTS

4.5.1 Setup

We implement all four models with Sentence-Transformers. Each trains for one epoch with batch size 16, maximum length 2048, and learning rate 3×10^{-4} . Although a single epoch may appear short, the dataset is large and both late-fusion and dual models plateau during that epoch; additional training produced no significant gains. The comparison therefore measures the architectures under the same compute budget rather than assuming that the dual models are fully optimized in

an unconstrained setting. Fine-tuning Qwen3-0.6B uses eight A100 GPUs for approximately two hours, an estimated \$22.40 at \$1.40 per GPU-hour. The speech models train for approximately eight hours on one H100. They are cheaper because the large text model is frozen: late fusion updates only its small speech aligner, whereas the dual architecture updates the speech encoder and projection modules.

4.5.2 Document Retrieval

We evaluate normalized discounted cumulative gain (nDCG) at ranks 5 and 10. NLLB-LLM2Vec [91] is an 8B-parameter text-only baseline with fixed 4096-dimensional embeddings. A pipelined baseline first transcribes speech and then uses our text-only model.

Approach	4096		1024		512		256		128	
	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10
NLLB-LLM2Vec	57.98	61.53	–	–	–	–	–	–	–	–
Late-Fusion	–	–	69.85	74.49	66.86	71.04	61.58	67.05	56.13	62.30
Pipelined	–	–	57.09	62.82	54.07	59.14	45.60	51.56	41.21	47.34
Dual-Retrieval	–	–	46.96	53.70	45.27	52.48	41.97	49.78	38.47	44.40
Dual-Alignment	–	–	41.56	47.42	41.02	46.76	38.24	44.18	35.47	40.47

Table 4.5: Document-retrieval nDCG on Kallaama-Retrieval-Eval.

Approach	4096		1024		512		256		128	
	@5	@10	@5	@10	@5	@10	@5	@10	@5	@10
NLLB-LLM2Vec	55.98	59.43	–	–	–	–	–	–	–	–
Late-Fusion	–	–	57.89	61.19	55.03	58.90	50.87	54.43	41.56	46.60
Dual-Retrieval	–	–	41.28	45.54	40.18	45.06	39.84	45.17	39.24	44.34
Dual-Alignment	–	–	38.07	41.82	37.33	41.69	38.04	41.29	34.46	38.53

Table 4.6: Document-retrieval nDCG on Fleurs-Retrieval-Eval.

Tables 4.5 and 4.6 show that late fusion performs best. On Kallaama, it outperforms NLLB-LLM2Vec despite being ten times smaller with embeddings four times shorter. It also outperforms the ASR pipeline. The dimension gap is larger on FLEURS, suggesting larger representations are more robust to low-quality speech.

The pipelined baseline already includes the first major source of cascading error: speech is transcribed before its text representation is retrieved. Adding machine translation would introduce another imperfect intermediate prediction, as well as additional latency, before retrieval. An ASR–translation–retrieval cascade should therefore be regarded as a less favorable operating point, especially for Wolof, where intermediate ASR and translation models are weaker than their high-resource counterparts. This expectation is an inference from

the measured ASR–retrieval result rather than a directly measured three-stage baseline.

Finding 1: Late fusion outperforms dual architectures for cross-modal adaptation

Late fusion provides stronger cross-modal generalization than dual encoders while training fewer parameters.

4.5.3 Speech Keyword Spotting

We evaluate Urban Bus [29], retrieving the station or landmark spoken by the user. This is equivalent to transcription retrieval.

Approach		1024	512	256	128
Late-Fusion	F1	88.79	84.85	79.80	76.86
	Recall	89.79	86.49	81.98	79.88
Dual-Retrieval	F1	68.64	58.33	50.86	44.59
	Recall	71.17	62.76	56.76	52.25
Dual-Alignment	F1	76.07	67.67	67.67	65.22
	Recall	77.78	70.87	70.87	68.47

Table 4.7: Keyword-spotting performance across embedding dimensions.

Late fusion performs best at every dimension. Dual encoders nevertheless perform much better here than on document retrieval, and query alignment outperforms direct retrieval training.

Finding 2: Dual models remain useful for less semantically demanding tasks

Dual encoders remain viable for transcription retrieval, which requires less semantic understanding than document retrieval.

4.5.4 Generalization to an Unseen Task

We evaluate speech intent detection on WolBanking77 [56], which has ten banking intents. Zero-shot classification selects the label most similar to a speech request. In the n -shot setting, we follow Set-Fit [108]: contrastive fine-tuning improves representations before a logistic-regression head is trained.

Zero-shot F1 is substantially above the 10% random baseline and reaches 96.11 with 16 examples per class. Figure 4.7 shows that larger embeddings adapt faster, while smaller ones catch up as data increases.

<i>n</i> -shot	F1	Recall
0	44.79	50.64
1	56.29	58.50
2	60.40	61.93
4	87.18	88.46
8	92.54	92.51
16	96.11	96.10

Table 4.8: Few-shot speech intent detection at dimension 1024.

Finding 3: Large dimensions adapt quickly; smaller dimensions need more data

Large dimensions adapt rapidly to new tasks; smaller dimensions require more examples but eventually approach their performance.

4.6 ANALYSIS

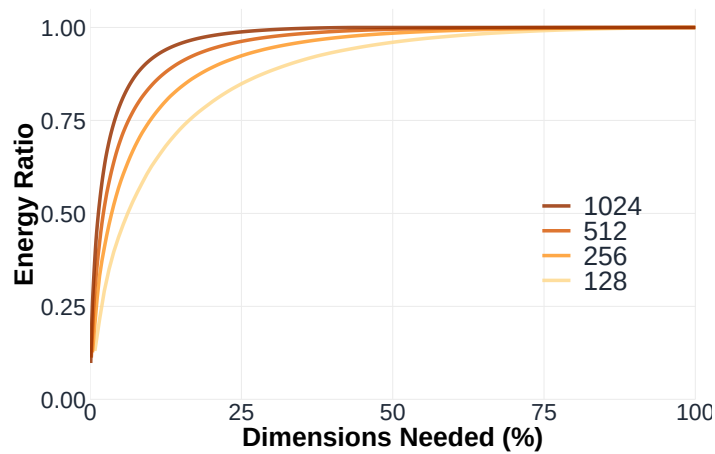


Figure 4.5: Dimensions needed for a given energy ratio.

4.6.1 Speech Quality in FLEURS

FLEURS asks speakers to read translated texts, but Wolof’s standardized writing system is not widely taught. Its hesitations thus primarily reflect difficulty reading Wolof rather than the discourse hesitations that naturally occur in spontaneous conversation. Compared with Kallaama, FLEURS contains more hesitations and lower volume. It averages 7.51 characters per second versus 16.88 for Kallaama, while

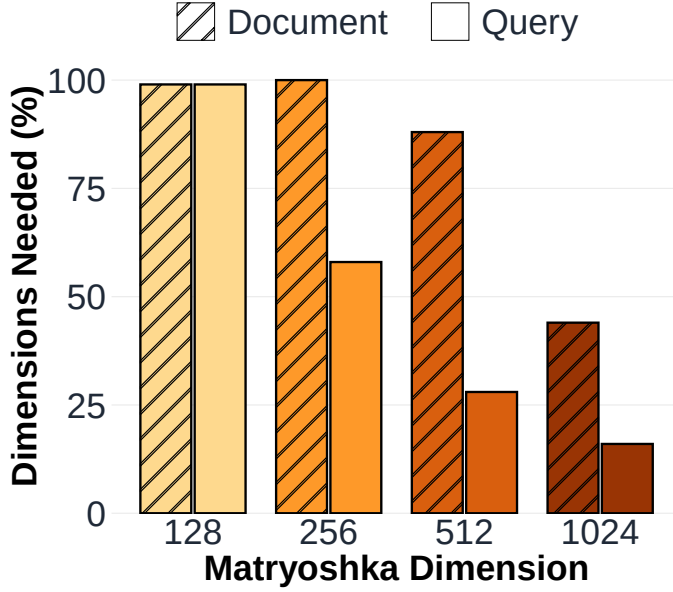


Figure 4.6: Dimensions needed for full energy.

mean loudness is -49.01 dB versus -23.81 dB. This helps explain Table 4.6 and the benefit of larger dimensions for low-quality speech.

Finding 4: Wolof speech in FLEURS is unnatural

Read Wolof in FLEURS is less natural, more hesitant, and quieter than spontaneous Kallaama speech.

4.6.2 Prompting

Without prompts, late fusion obtains $67.07/71.68$ nDCG@5/10 on Kallaama at dimension 1024, compared with $46.96/53.70$ for the dual model. Its advantage is therefore not explained only by prompt support. Task-aligned prompts still help: the document prompt produces the best document scores ($68.85/74.49$), while the transcription prompt produces the best keyword scores (88.79 F1, 89.79 recall).

4.6.3 Rank of Matryoshka Dimensions

We compute the cumulative energy ratio

$$R(k) = \frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^d \lambda_j}, \quad (4.4)$$

where the λ_i are descending eigenvalues of the representation covariance matrix on Kallaama. Figure 4.5 shows that larger embeddings reach full energy using a smaller fraction of coordinates and are more

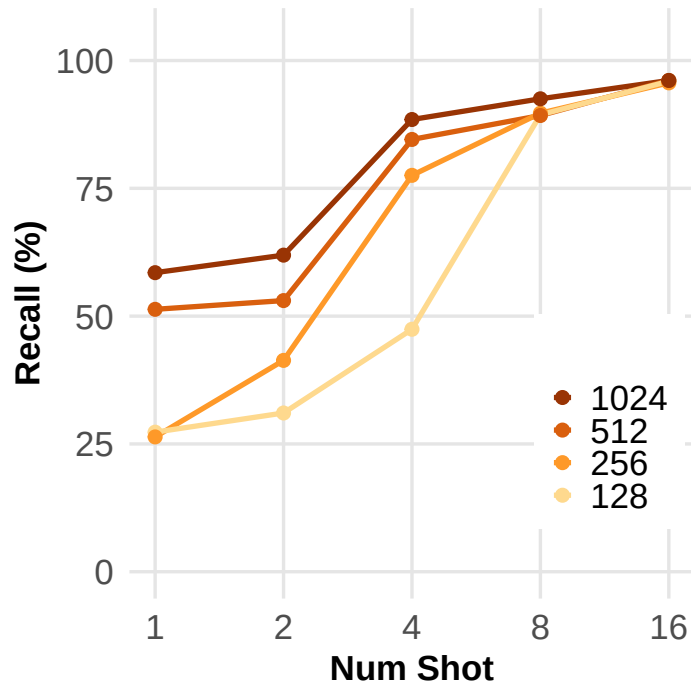


Figure 4.7: Recall by dimension and examples.

low-rank. Yet smaller prefix embeddings do not match their accuracy, suggesting slicing discards critical information. Prior work similarly identifies gradient variance and rigid slicing as MRL limitations [111, 122]. Figure 4.6 also shows that documents have higher ranks than speech queries; dimensions 128 and 256 are full-rank.

Finding 5: Information is distributed unevenly across Matryoshka dimensions

Large representations concentrate information in relatively few components, leaving scope for better compression than prefix slicing.

4.6.4 Deployment Costs

We index float16 embeddings with chromadb and measure indexing throughput, disk usage, and single-result query latency. Figure 4.8 shows diminishing returns: larger embeddings provide smaller accuracy gains while increasing storage. Smaller embeddings accelerate retrieval substantially, although indexing speed improves more gradually. The best dimension depends on the task and compute budget.

The stronger degradation of small dimensions on FLEURS does not imply that low-dimensional embeddings work only with clean or near-perfect recordings. It is the standard efficiency–accuracy tradeoff

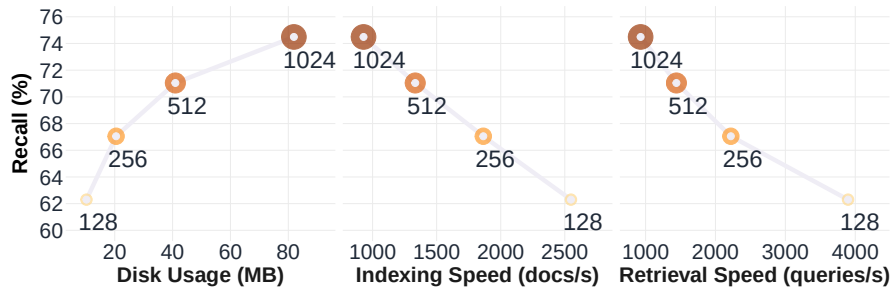


Figure 4.8: Indexing, storage, query-time, and accuracy tradeoffs across dimensions.

becoming steeper under more challenging acoustic conditions: larger vectors preserve more information and are more robust, while smaller vectors retain their storage and speed advantages at the cost of a larger accuracy drop. Deployment should therefore select dimension jointly with expected input quality rather than treating one dimension as universally optimal.

4.7 LIMITATIONS

SYNTHETIC DATA. Most documents are generated from transcribed and translated speech. This enables scale but may not reproduce real-world diversity and noise. It also assumes usable ASR and translation systems, which many languages lack.

BEYOND MATRYOSHKA SLICING. Only a small subset of coordinates represents most information, suggesting prefix slicing is suboptimal. Sparse-coding alternatives [111] or dynamic structured sparsity may offer more flexible compression.

4.8 CONCLUSION

We introduced cross-lingual speech–text Matryoshka embedding models that retrieve French documents directly from Wolof speech. Late modality fusion consistently outperforms dual encoders and an ASR pipeline, while the learned representations transfer to keyword spotting and intent detection.

The dimension and rank analyses expose a central limitation: information is concentrated in a small number of components, yet prefix-truncated embeddings do not preserve enough of it to match the largest representation. This connects adaptive representation learning to activation sparsity and motivates more flexible mechanisms for selecting informative dimensions at inference time.

Part IV

APPENDIX



APPENDIX TEST

Lorem ipsum at nusquam appellantur his, ut eos erant homero concludaturque. Albucius appellantur deterruisset id eam, vivendum partiendo dissentiet ei ius. Vis melius facilisis ea, sea id convenire referrentur, takimata adolescens ex duo. Ei harum argumentum per. Eam vidit exerci appetere ad, ut vel zzril intellegam interpretaris.

More dummy text.

A.1 APPENDIX SECTION TEST

Test: [Table A.1](#) (This reference should have a lowercase, small caps A if the option floatperchapter is activated, just as in the table itself → however, this does not work at the moment.)

LABITUR BONORUM PRI NO	QUE VISTA	HUMAN
fastidii ea ius	germano	demonstratea
suscipit instructor	titulo	personas
quaestio philosophia	facto	demonstrated

Table A.1: Autem usu id.

A.2 ANOTHER APPENDIX SECTION TEST

Equidem detraxit cu nam, vix eu delenit periculis. Eos ut vero constituto, no vidit propriae complectitur sea. Diceret nonummy in has, no qui eligendi recteque consetetur. Mel eu dictas suscipiantur, et sed placerat oporteat. At ipsum electram mei, ad aequae atomorum mea. There is also a useless Pascal listing below: [Listing A.1](#).

Listing A.1: A floating example (listings manual)

```
for i:=maxint downto 0 do
begin
{ do nothing }
end;
```

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DECLARATION

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Saarbrücken, February 2025

André Miede & Ivo Pletikosić

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